

SET THEORY

1. SET

Definition Any well defined collection, class or list of distinct objects is called a set. The objects in a set can be numbers, people, letters, rivers, countries etc. The phrase 'well defined' signifies that we can decide with certainty whether a particular object belongs or does not belong to the set. A set is any collection of distinct and distinguishable objects.

The objects in a set are called the *elements* or *members* of the set.

Examples of set:

1. The set of numbers 4, 6, 8, 10.
2. The set of vowels a, e, i, o, u.
3. The set of men Sohan, Mohan, Surendra and Suresh.
4. The set of rivers Satlej, Ravi and Jhelum.
5. The set of roots of the equation $x^2 - 7x - 12 = 0$.
6. The set of citizens of India.
7. The set of Indian students who have gone abroad.
8. The set of districts in Punjab state.

The sets given in the first four examples are well defined and every elements of the set is known. The members of sets given in examples 5, 6, 7 and 8 have not been written directly, but the rules to find out the members of those sets have been given. Such rules decide whether or not a particular object is a member of the set. Hence these sets are also well defined.

Notation: Sets are usually denoted by capital letters or upper case letters A, B, C, X, Y, ..., and the elements in a set are denoted by lower case letters a, b, c, x, y, ...

2. REPRESENTATION OF SETS

There are two ways of writing down a set, tabular form and set builder form.

(1) **Tabular form or Roaster form of a set:** In this form all the members of the set are listed. These members are separated by commas and enclosed in braces { }.

Example: If the elements of a set A are 2, 5, 7, 8, 11, then it can be written in tabular form as $A = \{2, 5, 7, 8, 11\}$. Representation of the set in this way is known as tabular form of a set.

(2) **Set-builder form or Rule form of a set:** In this form a set is defined by stating the properties which are satisfied by its elements. Here the rule governing the membership of the set is stated.

Example: A set of all even positive integers is written in set-builder form as

$$B = \{ x : x \text{ is even positive integer} \}$$

or $B = \{ x \mid x \text{ is even positive integer} \}$

and this is read "B" is a set of numbers x such that x is an even positive integer.

Example 1: Express the set whose elements are 3 and 4 in both the forms.

Solution: (1) Tabular form

$$A = \{3, 4\}$$

(2) Set-builder form

$$A = \{x : x^2 - 7x + 12 = 0\}$$

Example 2: Write down the set $S = \{x : x^2 - 8x + 15 = 0\}$ in tabular form and the set $A = \{\pm 1, \pm 3, \pm 5, \pm 7, \dots\}$ in set builder form.

Solution: $S = \{3, 5\}$ and $A = \{x : x \text{ is an odd integer}\}$

Note: If x is a member of a set A , then we write $x \in A$, which is read "x belongs to A or x is contained in A or x is an element or member of A."

If x is not a member of a set A , then we write $x \notin A$.

If x and y are both members of a set A , then we write $x, y \in A$. In example: $2, 3 \in A, 3, 5 \in A, 6 \notin A$

3. CARDINALITY OR CARDINAL NUMBER OR DIMENSION OR ORDER OF A SET AND EQUIVALENT SETS

The number of distinct elements in a set A is also called the number or cardinal number of the set and is denoted as $n(A)$ or $O(A)$ or $|A|$.

If $n(A) = n(B)$, A & B are said to be equivalent sets.

Example: If $S = \{3, 5, 7, 9, 11\}$, then the cardinal number of this set is 5 because there are only five elements in this set and we write $n(S) = |S| = 5$.

Note: $\{0\}$ and $\{\phi\}$ are equivalent sets each with cardinality equal to one.

4. FINITE AND INFINITE SETS

A set is called finite if it has n distinct elements where $n \in \mathbb{N}$. Such a set consists of a specific number of different elements. In such a set the process of counting different members comes to an end. Otherwise a set is infinite. We give below examples of such sets.

(1) A set of positive integers greater than 4 and less than 11 is finite because its six elements 5, 6, 7, 8, 9, 10 can be counted.

(2) $A = \{2, 4, 6, \dots\}$ is an infinite set.

(3) $B = \{x : x \text{ is an Indian citizen}\}$ is a finite set.

Note 1. The definitions of finite and infinite sets given earlier are general.

2. An empty set is regarded as a finite set.

5. EQUALITY OF SETS

If every member of set A is a member of set B and every member of set B is also a member of set A , then two sets A and B are said to be equal and we write $A = B$.

We give below examples of equal sets.

(1) If $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 1, 2, 4, 5\}$, then $A = B$, because every member of set A is a member of set B and every member of set B is a member of set A .

Note: It is evident from here that a set does not change if its elements are rearranged.

(2) If $A = \{6, 9, 9, 7\}$ and $B = \{7, 6, 7, 9\}$, then $A = B$ as every member of A is a member of B and every member of B is a member of A . Also if $C = \{6, 7, 9\}$, then $C = A = B$.

Note: It is evident from this example that a set does not change if its elements are repeated.

(3) If $A = \{3, 4\}$, $B = \{3, 4, 4, 3\}$, $C = \left\{3, \frac{12}{3}, \frac{6}{2}\right\}$ and $D = \{x : x^2 - 7x + 12 = 0\}$, then $A = B = C = D$.

Generally a set is written without repeating its elements.

6. NULL SET OR VOID SET OR EMPTY SET

A set which contains no element is called a Null set. It is denoted by the symbol $\emptyset$ or $\{\}$. $\{\}$ is the notation of null set in Roaster form.

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Examples of Null set: The following are the examples of null set:

- (1) The set of integers between $9/4$ and $11/4$.
- (2) The set of integers that are less than 4 and greater than 8.
- (3) The set $A = \{x \mid x^2 = 16 \text{ and } x \text{ is odd}\}$
- (4) The set of persons of age greater than 400 years.
- (5) The set of married bachelors.
- (6) The set $\{x : x \text{ is a real number and } x^2 = -1\} = \phi$ as there is no real number whose squares is negative.

Note: (1) The symbol $\emptyset$ indicating a null set is not enclosed within braces.

(2) The set $A = \{0\}$ is not a null set because it contains one element 0.

(3) A null set is regarded as a finite set.

(4) $\{\phi\}$ is not a null set.

7. SINGLETON SET

A set containing only one element is called a singleton set, e.g. $A = \{3\}$ and $B = \{0\}$ are both singleton sets.

8. SUBSETS

If every member in a set A is also a member of set B, then A is said to be a subset of B or contained in B. We write this relation as $A \subseteq B$.

There may or may not be elements in B that do not belong to A.

Examples of Subsets

(1) If $A = \{2, 4, 6\}$ and $B = \{3, 2, 4, 5, 6\}$ then A is a subset of B because every member of A is also a member of B.

(2) If $A = \{1, 3, 5\}$ and $B = \{5, 3, 1\}$, then A is subset of B and also B is a subset of A. Here $A = B$.

Note: It is evident from this example that every set is a subset of itself i.e. $A \subseteq A$.

(3) If $A = \{2, 4, 6, 8, \dots\}$ and $B = \{2, 4, 8, 16, 32, \dots\}$, then B is a subset of A.

(4) Let A be a set and $B = \{A, \{A\}\}$. Then $A \in B$ and $\{A\} \in B$. Therefore $\{A\} \subseteq B$ and $\{\{A\}\} \subseteq B$. But it is wrong to say that $A \subseteq B$.

(5) Given $S = \{2, a, \{3\}, 4\}$ and $R = \{\{a\}, 3, 4, 1\}$ then

(i) $a \in S$ (ii) $\{a\} \in R$ (iii) $\{a, 4, \{3\}\} \subseteq S$ (iv) $\{\{a\}, 1, 3, 4\} \subseteq R$ (v) $R \neq S$ (vi) $\phi \subset R$

9. EQUALITY OF SET AND SUBSETS

If A and B are two sets, then $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$ or $A = B \Leftrightarrow A \subseteq B$ and $B \subseteq A$. It means that every element of A belongs to B and every element of B belongs to A.

10. SUPERSET

If A is a subset of B, then B is said to be the superset of A. This relation is written as $B \supset A$ which reads "B contains A or B is superset of A".

If A is not a subset of B then it is written as $A \not\subset B$ or $B \not\supset A$.

Note: 1. The null set $\emptyset$ is regarded as a subset of every set.

2. If $A \not\subset B$ then there is at least one element in A that is not a member of B.

11. PROPER SUBSET

If B is a subset of A and B is not equal to A i.e. there exists at least an element in A which does not belong to B then B is said to be a proper subset of A.

Note: There is a distinction in the symbols used for "subset" and "proper subset". If B is a subset of A , they write it symbolically as $B \subseteq A$ and if B is a proper subset it is written as $B \subset A$.

For example:

If $A = \{1, 3, 4, 5, 6\}$ and $B = \{3, 6, 5\}$, then $B \subseteq A$ and $B \neq A$, or $A \not\subseteq B$ as $1, 4 \in A$ but $\notin B$. So it is written as $B \subset A$.

i.e. B is a proper subset of A .

Note: The null set $\emptyset$ is a proper subset of every other set except itself.

12. COMPARABILITY

Two sets A and B are said to be comparable if $A \subset B$ or $B \subset A$ or $A = B$

i.e. if one of the two set is a subset of the other set.

A and B are not comparable if $A \not\subset B$ and $B \not\subset A$ and $A \neq B$

In this case there will be at least one element in each set which is not a member of the other set.

Examples of comparable and not comparable sets.

- (1) If $A = \{x, y, z\}$ and $B = \{a, y, x, b, z\}$ then A and B are comparable as A is a sub-set of B .
- (2) If $A = \{2, 3\}$ and $B = \{3, 4, 5\}$, then A and B are not comparable because neither A is a sub set of B nor is B a subset of A .

13. DISJOINT SETS

If no element of A is a member of B and no element of B is a member of A , then A and B are said to be disjoint sets. i.e. $A \not\subset B$, $B \not\subset A$ and $A \neq B$

In such a case the two sets A and B have no elements in common.

Example of disjoint sets. The following pairs of sets A and B are disjoint sets.

- (1) $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$.
- (2) $A = \{x : x \text{ is positive}\}$ and $B = \{x : x \text{ is negative}\}$.
- (3) $A = \{x : x \text{ is an Indian man}\}$.

and $B = \{x : x \text{ is an Indian woman}\}$.

Note: The sets A and B are neither comparable nor disjoint where $A = \{2, 3, 7\}$ and $B = \{1, 3, 4, 6\}$

14. SET OF SETS OR FAMILY OF SETS

It is a set of elements which are sets themselves. For example the set $A = \{\{2, 3\}, \{3\}, \{5, 6, 7\}, \emptyset\}$ is a set whose members $\{2, 3\}$, $\{3\}$, $\{5, 6, 7\}$ and $\emptyset$ are sets themselves. We say that A is a set of sets or family of sets or class of sets.

$A = \{\{1, 3, 5\}, \{2, 4, 6, 8\}, \{9\}\}$ is also a family of sets.

Note: Set $A = \{\{1, 3\}, 7, \{3, 4, 5\}, 6\}$ is not a family of sets because some of its members i.e. 7 and 6 are not sets themselves.

15. UNIVERSAL SET OR UNIVERSE OF DISCOURSE

The set of which all sets under investigation are sub-sets is called Universal set. The universal set is generally denoted by U .

Example of Universal Set

1. All the person in the world constitute the universal set for the purpose of population studies.
2. In the study of plane geometry, all the points in the plane constitute a universal set.

16. POWER SET

The set containing all the sub-sets of a given set A as its elements is called a power set of A and is denoted by 2^A or $P(A)$.

or, $P(S) = \{T : T \subseteq S\}$, $\emptyset$ and S both $\in P(S)$, where S is any set.

Examples of power sets:

1. If $A = \{3, 4\}$, then all the sub-sets of A are $\{3, 4\}$, $\{3\}$, $\{4\}$, $\emptyset$
Hence its power set is $2^A = \{\{3, 4\}, \{3\}, \{4\}, \emptyset\}$
2. If $B = \{a, b, c\}$, then its power set is
 $2^B = \{\{a, b, c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a\}, \{b\}, \{c\}, \emptyset\}$.

17. NUMBER OF ELEMENT IN POWER SET

If the number of elements in any set is n then the number of elements in its power set is

$${}^nC_n + {}^nC_{n-1} + {}^nC_{n-2} + \dots + {}^nC_1 + 1 = 2^n.$$

Example 3: The number of elements in two finite sets A and B are n_1 and n_2 . The total number of subsets of the first set (i.e., number of elements in power set of A) is eight times the total number of subsets of the second set (i.e., number of elements in power set of B) Find value of $n_1 - n_2$

Solution: $n(A) = n_1$, $n(B) = n_2$

Total number of subsets of $A = 2^{n_1}$

Total number of subsets of $B = 2^{n_2}$

As per condition given in the question

$$2^{n_1} = 8 \cdot 2^{n_2}$$

$$\text{or, } \frac{2^{n_1}}{2^{n_2}} = 8 \text{ or } 2^{n_1 - n_2} = 8 = 2^3$$

$$\text{or, } n_1 - n_2 = 3$$

18. SOME THEOREMS

Theorem 1: If $A \subset B$ and $B \subset A$, then

$$A = B$$

Proof: $A \subset B$ implies that every element of A belongs to B and $B \subset A$ implies that every element of B belongs to A .

So A and B have the same elements.

$$\therefore A = B$$

19. SET OPERATIONS

Like the processes of addition, subtraction, multiplication and division in the number systems, there are certain fundamental operations on sets which are described below:

(1) **Union of sets:** The Union of two sets A and B is the set of all elements which belong to A or to B or to both A and B . The union of A and B is denoted by $A \cup B$ and is read 'A union B'

The union of A and B can also be written as $A \cup B = \{x : x \in A \text{ or } x \in B\}$. Here the word 'or' is used in inclusive sense

Note: 1. $A \cup B$ and $B \cup A$ are the same set i.e. $A \cup B = B \cup A$

2. A as well B is always a sub-set of $A \cup B$ i.e. $A \subseteq (A \cup B)$ and $B \subseteq (A \cup B)$.

3. The set $A \cup B$ is also written as $A + B$ (set theoretic sum).

Example: If $A = \{1, 2, 3, 4\}$ and $B = \{5, 2, 4, 6\}$ then $A \cup B = \{1, 2, 3, 4, 5, 6\}$.

The easy way to write the union of two sets is first to write all the element of any one of the two sets and then include those elements of the other set that are not contained in the first one.

$A \cup \emptyset = \emptyset \cup A = A$, Hence $\emptyset$ is an identity element for the operation of union.

$$A \cup U = U.$$

(2) **Intersection of sets:** The intersection of two sets A and B is the set of elements which belong to A as well as to B and it is denoted by $A \cap B$.

It is the set of all elements which are common to A and B.

The intersection of A and B is also written as $A \cap B = \{x : x \in A \text{ and } x \in B\}$

$A \cap B$ is also written as AB and is called the set theoretic product.

Note: 1. By definition $(A \cap B) \subseteq A$ and $(A \cap B) \subseteq B$.

2. If A and B are disjoint sets, then $A \cap B = \emptyset$.

3. $A \cap \emptyset = \emptyset \cap A = \emptyset$

4. $A \cap U = U \cap A = A$

Hence U (universal set) is an identity for the operation of intersection of sets.

5. If $A \cap B \neq \emptyset$, then A and B are said to be intersecting or overlapping sets.

Examples of Intersection of sets:

(1) If $A = \{3, 4, 6, 8\}$ and $B = \{2, 6, 8, 4\}$ then $A \cap B = \{4, 6, 8\}$

(2) If $A = \{x \mid x = 3n, n \text{ is positive integer}\}$

and $B = \{x \mid x = 2n, n \text{ is positive integer}\}$

Then $A = \{x : x \text{ is a positive integer divisible by } 3\}$

and $B = \{x : x \text{ is a positive integer divisible by } 2\}$

Therefore $A \cap B = \{x : x \text{ is a positive integer divisible by } 3 \text{ and } 2\}$

$= \{x : x \text{ is a positive integer divisible by } 6\}$

$= \{x : x = 6n, n \text{ is a positive integer}\}$

(3) If $A = \{p, q, r, s, t\}$ and $B = \{a, b, c, d\}$ then $A \cap B = \emptyset$

The operation of union and intersection can also be defined for three or more sets in a similar manner.

Thus $A \cup B \cup C = \{x \mid x \in A \text{ or } x \in B \text{ or } x \in C\}$

and $A \cap B \cap C = \{x \mid x \in A \text{ and } x \in B \text{ and } x \in C\}$

In general $A_1 \cup A_2 \dots \cup A_n$ is denoted by $\bigcup_{k=1}^n A_k$ and $A_1 \cap A_2 \dots \cap A_n$ is denoted by $\bigcap_{k=1}^n A_k$.

(3) **Difference of Sets:** The difference of two sets A and B is the set of elements which belong to set A but do not belong to set B. This is denoted by $A - B$ or $A \sim B$ or $A \setminus B$ and is read "A difference B or A minus B".

$A - B$ can be written as $A - B = \{x \mid x \in A, x \notin B\}$. Similarly $B \sim A = \{x \mid x \in B \text{ and } x \notin A\}$

It is also called the complement of B with respect to A

$A - \emptyset = A, \emptyset - A = \emptyset, A - U = \emptyset$

Note: $A - B = A \cap B'$ and $B - A = B \cap A'$

It is evident by the definition of difference of two sets that:

(1) $(A - B) \subseteq A$.

(2) The sets $(A - B)$, $A \cap B$ and $B - A$ are mutually disjoint sets

Examples of difference of two sets :

(1) If $A = \{1, 3, 5, 7, 9\}$, and $B = \{3, 4, 5, 8\}$

then $A - B = \{1, 7, 9\}$ and $B - A = \{4, 8\}$

(3) If $A = \{x \mid x \text{ is a real number}\}$

and $B = \{x \mid x \text{ is a rational number}\}$

then $A - B = \{x \mid x \text{ is an irrational number}\}$ and $B - A = \emptyset$

(4) **Complement of a set:** The complement of a set A is the set of elements which do not belong to A. It is, therefore, the difference of the universal set and the set A. The complement of a set A is denoted by A' or A^c .

Examples of Complement of a set :

(1) If $A = \{x \mid x \text{ is odd natural number}\}$

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and $U = \{x \mid x \text{ is natural number}\}$

then $A' = \{x \mid x \text{ is even natural number}\}$.

(2) If $U = \{1, 2, 3, \dots, 9, 10\}$

and $A = \{2, 4, 6, 8, 10\}$

then $A' = U - A$

$$= \{x \mid x \in U, x \notin A\}$$

$$= \{1, 3, 5, 7, 9\}$$

(3) If $U = \{x \mid x \text{ is a letter of English language}\}$

and $A = \{u, v, w, x, y, z\}$

then $A' = \{a, b, c, \dots, s, t\}$

(4) If $A = \{x : x \text{ is integer and } x \leq 7\}$

and $U = \mathbb{Z}$ (set of integers), then

$$A' \text{ or } \bar{A} = \{x : x \text{ is an integer and } x > 7\}$$

Some properties associated with complement of a set.

$$(a) A \cup A' = U \quad (b) A \cap A' = \emptyset \quad (c) U' = \emptyset \text{ and } \emptyset' = U \quad (d) (A')' = A$$

$$(e) A - B = A \cap B'$$

Proof: $A - B = \{x \mid x \in A, x \notin B\}$

$$= \{x \mid x \in A, x \in B'\}$$

$$= A \cap B'$$

(5) Symmetric difference of two sets: Symmetric difference of two sets A and B is given by

$$A \oplus B \text{ or } A \triangle B = (A - B) \cup (B - A)$$

Similarly, $B \oplus A \text{ or } B \triangle A = (B - A) \cup (A - B)$,

$$\text{Hence } A \oplus B = B \oplus A$$

It is the set of all elements that belong to A or to B but not to both A and B. Thus

$$A \oplus B = \{x : (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)\}$$

(6) Min set: Let P and Q be two subsets of a universal set U. Then all possible sets that are intersection of P, P', Q and Q' (except $P \cap P'$ and $Q \cap Q'$) are called min sets with respect to P and Q. These are $P \cap Q$, $P \cap Q'$, $P' \cap Q$ and $P' \cap Q'$.

(7) Max set: Let P and Q be two subsets of a universal set U. Then all possible sets that are union of P, P', Q and Q' (except $P \cup P'$ and $Q \cup Q'$) are called max set with respect to P and Q. These are

$$P \cup Q, P' \cup Q, P \cup Q' \text{ and } P' \cup Q'$$

Note: Dash (') denotes the complementation

Example 4: If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$P = \{1, 3, 5, 7\}$, $Q = \{4, 6, 8, 10\}$ find min sets and max sets

Solution: $P' = \{2, 4, 6, 8, 9, 10\}$, $Q' = \{1, 2, 3, 5, 7, 9\}$

Min sets are:

$$(a) P \cap Q = \emptyset$$

$$(b) P' \cap Q = \{4, 6, 8, 10\}$$

$$(c) P \cap Q' = \{1, 3, 5, 7\}$$

$$(d) P' \cap Q' = \{2, 9\}$$

Max sets are:

$$(a) P \cup Q = \{1, 3, 4, 5, 6, 7, 8, 10\}$$

$$(b) P' \cup Q = \{2, 4, 6, 8, 9, 10\}$$

$$(c) P \cup Q' = \{1, 2, 3, 5, 7, 9\}$$

$$(d) P' \cup Q' = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

20. ALGEBRA OF SETS

Laws of Algebra of Sets:

(that are satisfied by the operations of Union, Intersection and Complement)

Idempotent laws

1. a. $A \cup A = A$
b. $A \cap A = A$

Associative laws

2. a. $(A \cup B) \cup C = A \cup (B \cup C)$
b. $(A \cap B) \cap C = A \cap (B \cap C)$

Commutative laws

3. a. $A \cup B = B \cup A$
b. $A \cap B = B \cap A$

Distributive laws

4. a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Identity laws

5. a. $A \cup \emptyset = A$
b. $A \cap U = A$

Domination laws

6. a. $A \cup U = U$
b. $A \cap \emptyset = \emptyset$

Involution law

7. $(A')' = A$

Complement laws

8. a. $A \cup A' = U$
b. $A \cap A' = \emptyset$
9. a. $U' = \emptyset$
b. $\emptyset' = U$

De Morgan's law

10. a. $(A \cup B)' = A' \cap B'$
b. $(A \cap B)' = A' \cup B'$
11. a. $A - (B \cup C) = (A - B) \cap (A - C)$
b. $A - (B \cap C) = (A - B) \cup (A - C)$

Duality

If E is any identity of algebra of sets, then the dual E^* of E is the expression obtained by replacing each occurrence of $\cup, \cap, U$ and $\emptyset$ in E by $\cap, \cup, \emptyset$ and U respectively. [All b 's are duals of a 's in section 20]. It is known as principle of DUALITY. If E is true, then E^* is also true.

20.1 Other Important Results in Set Theory

1. Number of elements belonging to A or B or both $= n(A \cup B)$
2. Number of elements not belonging to either of the two $= n(A \cup B)'$
3. Number of elements belonging to both A and $B = n(A \cap B)$
4. $n(A \cup B) = n(A) + n(B) - n(A \cap B) = n(A) + n(B)$, if $n(A \cap B) = \emptyset$
[counting principle]
It can be extended to more than two sets.
5. $n(A - B) = n(A) - n(A \cap B) = \text{number of elements belonging to } A \text{ only}$
6. $n(B - A) = n(B) - n(A \cap B) = \text{number of elements belonging to } B \text{ only}$

[Fundamental]

7. $n(A \cup B) = n(A - B) + n(B - A) + n(A \cap B)$
8. $n(A' \cup B') = n(A') + n(B') - n(A' \cap B')$
 $n(S) - n(A \cap B) = n(B' - A) + n(A' - B) - n(A' \cap B')$
9. $n(A' \cap B') = n(S) - n(A \cup B)$ [Neither belong to A nor to B i.e. belonging to neither of the two]
10. (a) $A \cup B = \emptyset \Leftrightarrow A = \emptyset \text{ \& } B = \emptyset$
 (b) $A - B = \emptyset \Leftrightarrow A \subseteq B$
 (c) $A - B = A \cap B^c$
11. (a) $x \notin A \text{ and } x \notin B \Leftrightarrow x \notin A \cup B$
 (b) $x \notin A \text{ or } x \notin B \Leftrightarrow x \notin A \cap B$
 (c) $x \in A \text{ or } x \in B \Leftrightarrow x \in A \cup B$
 (d) $x \in A \text{ and } x \in B \Leftrightarrow x \in A \cap B$
12. $A \subseteq B \Leftrightarrow B^c \subseteq A^c$
13. $A \cap (B - C) = (A \cap B) - (A \cap C)$
14. $A - B = A \cap B'$ and $B - A = B \cap A'$
15. If $x \in (A \cap B)'$, then $x \notin A \cap B$
 i.e. $x \notin A$ or $x \notin B$
 $x \in A'$ or $x \in B'$
 $x \in A' \cup B'$
16. If $x \in A' \cup B'$, then $x \in A'$ or $x \in B'$
 $x \notin A$ or $x \notin B$
 $x \notin A \cap B$
 $x \in (A \cap B)'$

[Important]

SOLVED EXAMPLES

Example 5: If U is the universal set and A and B are two other sets, prove that

- (a) $(A \cap B) \subseteq A$ and $(A \cap B) \subseteq B$
 (b) $A \cap A = A$
 (c) $U \cap A = A$
 (d) $A \cap \emptyset = \emptyset$

Solution (a) Let $x \in (A \cap B)$

i.e. $x \in A$ and $x \in B$

$\therefore x \in (A \cap B)$ implies that $x \in A$ which is true for all $x \in A \cap B$ i.e. every element of $A \cap B$ belongs to A

$\therefore (A \cap B) \subseteq A$

Similarly we can show that $x \in (A \cap B) \Rightarrow x \in B$ i.e. $(A \cap B) \subseteq B$.

(b) $A \cap A$ is the set of elements which belong to A .

$\therefore A \cap A \subseteq A$

and as every element of A belongs to $A \cap A$

$\therefore A \subseteq A \cap A$

From (1) and (2)

$$A \cap A = A$$

(c) From Part (a), we know that

$$U \cap A \subseteq A$$

Let $x \in A$, then $x \in U$

$\therefore x \in (U \cap A)$

So $x \in A$ implies that $x \in (U \cap A)$, $\forall x \in A$

...(1)

...(2)

...(1)

$$\therefore A \subseteq U \cap A$$

from (1) and (2) $(U \cap A) = A$.

(d) We know that

$$(A \cap \emptyset) \subseteq \emptyset$$

But $\emptyset$ is a sub-set of every other set

$$\therefore \emptyset \subseteq (A \cap \emptyset)$$

from (1) and (2) $A \cap \emptyset = \emptyset$.

Example 6: If $U = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$X = \{2, 3, 6, 8\}$, $Y = \{4, 5, 7, 9\}$ and $Z = \{3, 5, 7\}$ find the following sets:

(a) X' (b) $(X \cup Y)'$ (c) $(Y - Z)'$

Solution: (a) $X' = \{x : x \in U, x \notin X\}$
 $= \{4, 5, 7, 9, 10\}$

(b) $X \cup Y = \{x : x \in X \text{ or } x \in Y\}$
 $= \{2, 3, 4, 5, 6, 7, 8, 9\}$

Then $(X \cup Y)' = \{x : x \in U, x \notin (X \cup Y)\} = \{10\}$

(c) $(Y - Z) = \{x : x \in Y, x \notin Z\} = \{4, 9\}$

Then $(Y - Z)' = \{x : x \in U, x \notin (Y - Z)\}$
 $= \{2, 3, 5, 6, 7, 8, 10\}$

Example 7: Prove that $Y - X' = Y \cap X$.

Solution: Let $x \in Y - X'$

Then $x \in Y$ and $x \notin X'$

or $x \in Y$ and $x \in X$

$\therefore x \in (Y \cap X)$,

Hence $x \in (Y - X')$ implies that $x \in (Y \cap X)$, $\forall x \in (Y - X')$.

$\therefore Y - X' \subseteq Y \cap X$

Again let $y \in Y \cap X$

then $y \in Y$ and $y \in X$

or $y \in Y$ and $y \notin X'$

or $y \in (Y - X')$

So $y \in Y \cap X$ implies that $y \in (Y - X')$, $\forall y \in Y \cap X$

$\therefore Y \cap X \subseteq Y - X'$

from (1) and (2) $Y - X' = Y \cap X$.

Example 8: Prove that $(X - Y) \cap Y = \emptyset$

Solution: Let $x \in (X - Y) \cap Y$,

i.e. $x \in (X - Y)$ and $x \in Y$

or $x \in X, x \notin Y$ and $x \in Y$.

Now the only set satisfying $x \in Y, x \notin Y$ is the null set.

$\therefore (X - Y) \cap Y = \emptyset$

Alternately: We see that $X - Y$ is set of elements belonging to X and not to Y . So $X - Y$ does not contain any element of Y , i.e. there is no elements common in $X - Y$ and Y except $\emptyset$.

$\therefore (X - Y) \cap Y = \emptyset$

Example 9: Prove that $B - A \subseteq A'$.

Solution: Let $x \in B - A$, i.e. $x \in B, x \notin A$

or $x \in B, x \in A'$.

SET THEORY

$\therefore x \in B - A$ implies that $x \in A'$
 $\therefore B - A \subseteq A'$.

Example 10: Prove that $(A \cup B) \cup C = A \cup (B \cup C)$

Solution: Let $S = (A \cup B) \cup C$ and $T = A \cup (B \cup C)$ we have to prove that $S = T$

Let $x \in S$, then $x \in (A \cup B)$ or $x \in C$

if $x \in (A \cup B)$, then $x \in A$ or $x \in B$.

Hence $x \in S$ implies that

$x \in A$ or $x \in B$ or $x \in C$ i.e. $x \in A$ or $x \in (B \cup C)$

i.e. $x \in A \cup (B \cup C)$

or $x \in T$.

$\therefore S \subseteq T$.

again let $y \in T$, then $y \in A$ or $y \in (B \cup C)$.

if $y \in (B \cup C)$, then $y \in B$ or $y \in C$,

Hence $y \in T$ implies that $y \in A$ or $y \in B$ or $y \in C$

i.e. $y \in (A \cup B)$ or $y \in C$

or $y \in (A \cup B) \cup C$

or $y \in S$.

$\therefore T \subseteq S$.

From (1) and (2)

or $(A \cup B) \cup C = A \cup (B \cup C)$.

Note: Similarly, we can prove that

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Example 11: Let $A = \{\phi, b\}$, construct the following sets.

- (i) $A - \phi$ (ii) $\{\phi\} - A$ (iii) $A \cup P(A)$ (iv) $A \cap P(A)$ (v) $\phi - A$

[UPTU., MCA., 2004-05]

Solution: Given that $A = \{\phi, b\}$ therefore

(i) $A - \phi = \{\phi, b\} - \phi = \{b\} = A$

(ii) $\{\phi\} - A = \{\phi\} - \{\phi, b\}$ set having elements of ϕ and not having elements ϕ and $b = \phi$

(iii) $A \cup P(A) = \{\phi, b\} \cup \{\phi, \{\phi\}, \{b\}, \{\phi, b\}\}$
 $= \{\phi, b, \{\phi\}, \{b\}, \{\phi, b\}\} = P(A)$

(iv) $A \cap P(A) = \{\phi, b\} \cap \{\phi, \{\phi\}, \{b\}, \{\phi, b\}\}$
 $= \{\phi, b\} = A$

(v) $\phi - A = \phi$

Example 12: Write sets (i) $\phi \cap \{\phi\}$ (ii) $\{\phi\} \cap \{\phi\}$ (iii) $\{\phi, \{\phi\}\} - \phi$.

Solution: (i) $\phi \cap \{\phi\}$ = sets of elements that are in ϕ as well as in $\{\phi\}$. But ϕ has no element and $\{\phi\}$ has one element ϕ . Therefore there is no element common to both the sets. Hence $\phi \cap \{\phi\}$ = set having no element = ϕ .

(ii) Similarly $\{\phi\} \cap \{\phi\}$ = set having elements common to both the sets = $\{\phi\}$.

(iii) $\{\phi, \{\phi\}\} - \phi$ = set having elements belonging to $\{\phi, \{\phi\}\}$ and not having elements belonging to $\phi = \{\phi, \{\phi\}\}$

Same Important Results

$\varnothing - \{\varnothing\} = \varnothing$	$\{\varnothing\} - \varnothing = \{\varnothing\}$
$\{a, b\} - \varnothing = \{a, b\}$	$\varnothing - \{a, b\} = \varnothing$
$\{a, b\} - \{\varnothing\} = \{a, b\}$	$\{\varnothing\} - \{a, b\} = \{\varnothing\}$
$\{\varnothing, b\} - \varnothing = \{\varnothing, b\}$	$\varnothing - \{\varnothing, b\} = \varnothing$
$\{\varnothing, b\} - \{\varnothing\} = \{b\}$	$\varnothing - \{\varnothing, b\} = \varnothing$ and $\{\varnothing\} - \{\varnothing, b\} = \varnothing$
$\{a, b\} \cup \varnothing = \{a, b\}$	$\{\varnothing\} \cap \{a, b\} = \varnothing$
$\{a, b\} \cup \{\varnothing\} = \{a, b, \varnothing\}$	$\varnothing \cap \{a, b\} = \varnothing$
$\{\varnothing, b\} \cup \varnothing = \{\varnothing, b\}$	$\varnothing \cap \{\varnothing, b\} = \varnothing$
$\{\varnothing, b\} \cup \{\varnothing\} = \{\varnothing, b\}$	$\{\varnothing\} \cap \{\varnothing, b\} = \{\varnothing\}$

2. VENN DIAGRAM

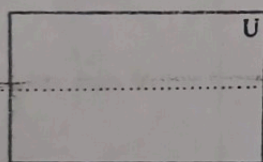
Geometrical figures (diagrams) used to represent sets are called Venn-Diagrams.

John Venn, an English Mathematician developed the idea of using geometrical figures to represent sets.

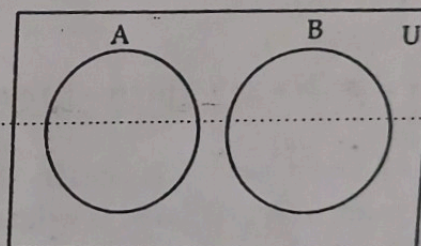
Another Mathematician Euler also used diagrams to represent sets and therefore these diagrams are also called Venn-Euler diagrams.

In a Venn diagram, generally, the universal set is represented by a rectangle and all other sets under consideration by bounded areas (generally, a circle) within the rectangle.

1. The adjacent diagram (i) shows a Universal set U.



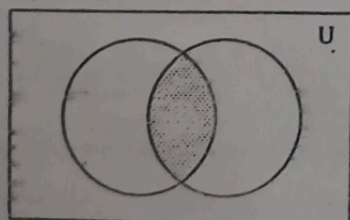
(i)



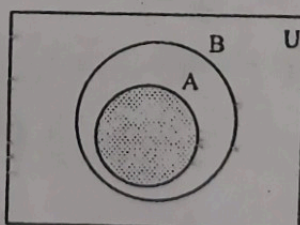
(ii)

2. The adjacent diagram (ii) shows two disjoint sets A and B, $A \cap B = \varnothing$.

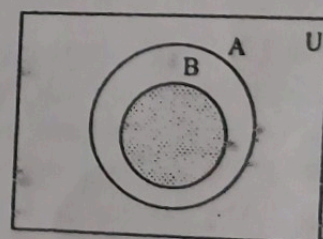
3. The following figures show that the sets A and B are overlapping sets and the shaded portion represents $A \cap B$.



(i)

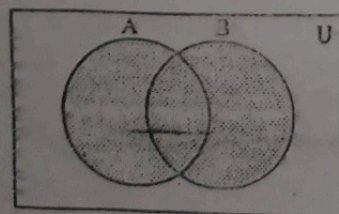


(ii)

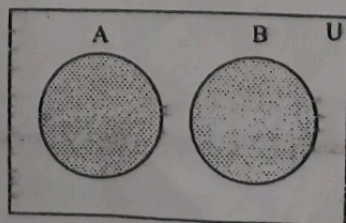


(iii)

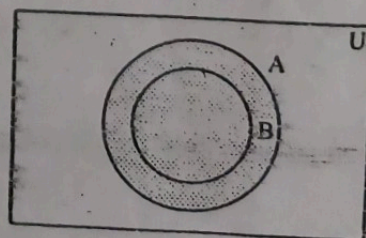
4. In the following figures the shaded portion represents $A \cup B$.



(i)

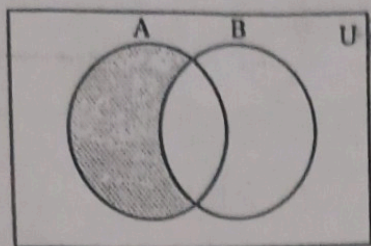


(ii)

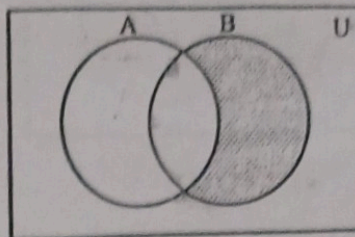


(iii)

5. Similarly, the shaded portion of the following Venn-diagrams represents $A - B$ and $B - A$

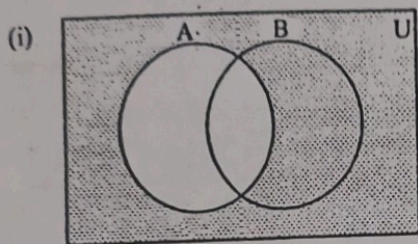


$(A - B)$

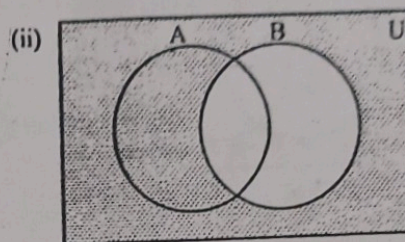


$(B - A)$

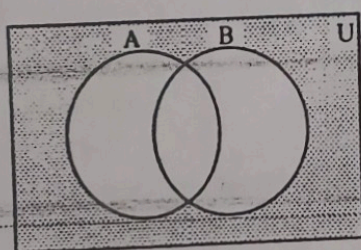
6. Other representations are given below:



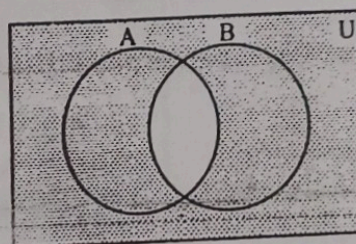
(A')



(B')

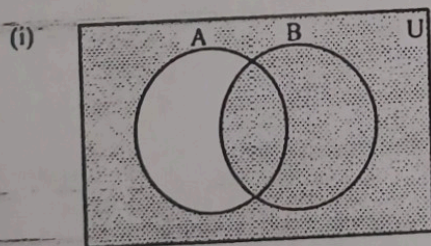


$U - (A \cup B)$ or $(A \cup B)'$

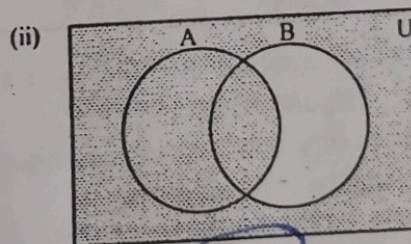


$(A \cap B)'$ or $U \sim (A \cap B)$

8.



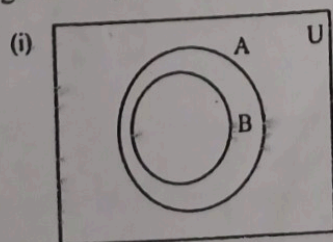
$(A - B)'$



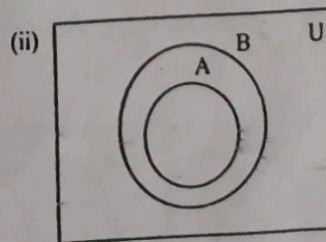
$(A - B)'$

$(B - A)'$

9. The following diagrams represent subsets,



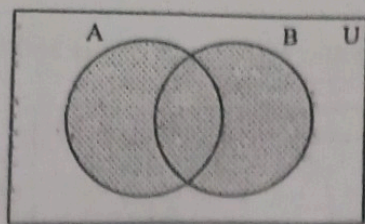
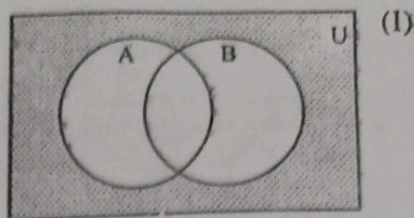
$(B \subset A)$



$(A \subset B)$

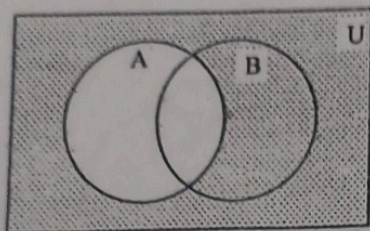
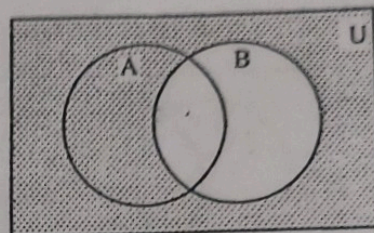
Example 13: Use Venn-diagrams to prove that $(A \cup B)' = A' \cap B'$.

Solution: $A \cup B$ = (The region which is in A or in B or in both).

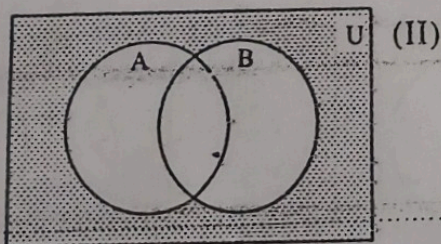
 $(A \cup B)$  $(A \cup B)'$

$(A \cup B)' =$ (The region in the universal set and not in $(A \cup B)$)

Now, $A' =$ (The region of universal set, which is not in A) and $B' =$ (The region of universal set, which is not in B).

 A'  B'

$A' \cap B' =$ (The region which is common to both A' and B')

 $A' \cap B'$

Since the shaded portions of (I) and (II) represent the same region.

Hence $(A \cup B)' = A' \cap B'$.

Example 14: Given universal set $U = \{a, b, c, d, e, f, g\}$

$A = \{b, c, d, e\}$ and $B = \{a, b, e\}$.

Draw Venn-diagram to represent the relationship between the given sets.

Use the diagram to find the following sets:

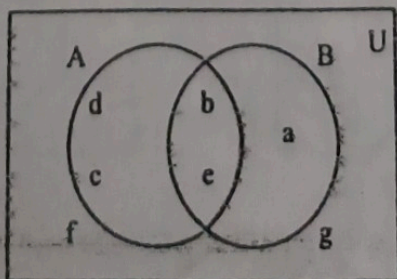
(i) A'

(ii) $A - B$

(iii) $B - A$

Solution: The given sets A and B are overlapping sets,
as $A \cap B = \{b, e\}$.

Therefore, the Venn-diagram representing the relationship between the given sets will be as shown alongside:



SET THEORY

- (i) $A' = \{\text{elements of universal set, which are not in } A\} = \{a, f, g\}$
 (ii) $A - B = \{\text{elements of } A, \text{ which are not in } B\} = \{c, d\}$
 (iii) $B - A = \{x : x \in B \text{ and } x \notin A\} = \{a\}$

(Ans.)
 (Ans.)
 (Ans.)

Example 15: In MCA Class of a certain college 42 students got distinction in Accounts, 60 got distinction in MFCS and 27 students in both the subjects.

Use Venn-diagram to find the number of students getting distinction.

- (i) In Accounts only
 (ii) In MFCS only;
 (iii) In Accounts or in MFCS or in both (i. e. total number of distinctions in both the subjects).

Solution: In the Venn-diagram drawn alongside; A represents the number of distinctions in Accounts, and M represents the number of distinctions in MFCS.

Since, the portion common to both the circles represents the number of distinctions in both the subjects, therefore, the number 27 is written in this common portion.

- (i) The region shaded by horizontal lines represents the number of distinctions in Accounts only.

$$\therefore 42 - 27 = 15 \text{ students got distinction in Accounts only}$$

- (ii) The region shaded by vertical lines represents the number of distinctions in MFCS only

$$\therefore 60 - 27 = 33 \text{ students got distinction in MFCS only.}$$

- (iii) It is clear from the figure that $15 + 27 + 33 = 75$ students got distinction either Accounts or in MFCS or in both.

Example 16: If $n(U) = 60$, $n(A) = 27$, $n(B) = 19$ and $n(A \cup B)' = 31$: Use Venn-diagram to find :

- (i) $n(A \cap B)$

- (ii) $n(A - B)$

Solution: Let $n(A \cap B) = x$

Since $n(A) = 27$,

$$\therefore n(A - B) = 27 - x \quad [n(A - B) = n(A) - n(A \cap B)]$$

Since $n(B) = 19$,

$$\therefore n(B - A) = 19 - x \quad [\text{since } (B - A) = n(B) - n(A \cap B)]$$

As shown in diagram, $n(A \cup B)' = 31$ is written outside A and B.

Since $n(U) = 60$,

$$\therefore 27 - x + x + 19 - x + 31 = 60,$$

$$\Rightarrow x = 17$$

Therefore, (i) $n(A \cap B) = x = 17$

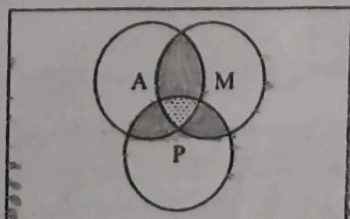
$$(ii) n(A - B) = 27 - x = 27 - 17 = 10$$

(Ans)

(Ans)

Example 17: If $A = \{\text{artists}\}$, $M = \{\text{Mathematicians}\}$, and $P = \{\text{Players}\}$, what is represented by the shaded area shown on the right ?

Solution: The shaded area corresponds to the intersection of the three sets and therefore includes those individuals who possess all



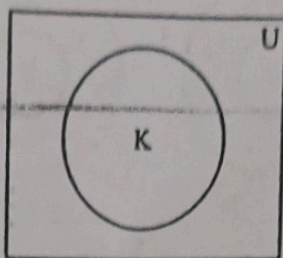
$n(A \cap M \cap P)$

the three qualities. Thus it represents the set of artists who are players as well mathematical.

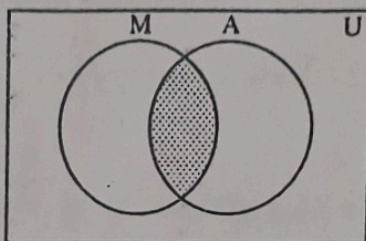
Example 18: Draw a Venn-diagram to illustrate the following statements :

- (a) All officers are men.
- (b) Some mathematical problems are geometrical problems.
- (c) Ajit is a Dwarf rich Indian.

Solution: (a) Let U = the set of all men; K = the set of all officers then the following diagram illustrates the given statement.

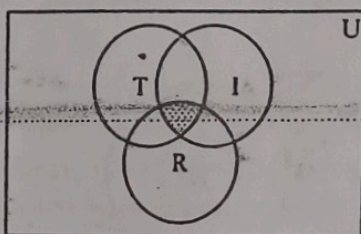


(b) Let U = { All problems }, M = { Mathematical problems } and A = { Geometrical problems }



Then the shaded region in figure represents the given statement.

(c) Let U = { All men }, T = { Dwarf men }, R = { Rich men } and I = { Indians }



Then the shaded region in figure represents the given statement and Ajit is an element in this area.

Example 19: The following advertisement appeared in a daily newspaper.

Required Mathematics teachers, who can teach Modern Algebra and can play Football. Apply before the 20th of this month to the director.

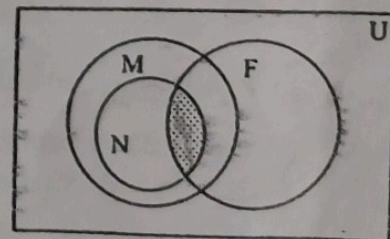
Taking the universal set U as the set of teachers, M as the set of mathematics teachers, N as the set of those who can teach modern algebra and F as the set of Players of football, draw a Venn-diagram to illustrate this. Shade the region which represents the teachers who meet all the requirements specified.

Solution: U = { Teachers }

M = { Maths teachers }

N = { Modern algebra teachers }

F = { Football players } The shaded area is shown in figure.



Example 20: In a class of 50 students, 30 study History, 25 study Economics and 11 study both the languages. Draw a Venn-diagram to represent it.

From the Venn-diagram find, how many students do not study any of these two subjects.

Solution: Let H represents a set of students who study History,

$\therefore n(H) = 30,$

SET THEORY

E represents a set of students who study Economics.

$$\therefore n(E) = 25.$$

Therefore, intersecting part of the two sets represents those students who study both, thus.

$$n(H \cap E) = 11.$$

The Venn-diagram will be as shown alongside.

Clearly number of students studying History only

$$\begin{aligned} &= n(H) - n(H \cap E) \\ &= 30 - 11 = 19 \end{aligned}$$

Number of students studying Economics only

$$\begin{aligned} &= n(E) - n(H \cap E) \\ &= 25 - 11 = 14 \end{aligned}$$

$\therefore$ Total number of students who study History or Economics or both

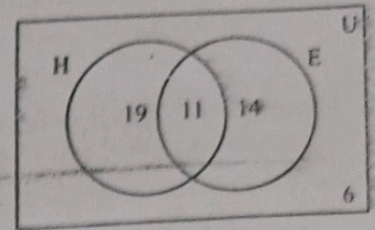
$$= 19 + 11 + 14 = 44$$

$$\text{i.e. } n(A \cup B) = 44$$

Number of students who do not study any of these two subjects

$$n(U) - n(A \cup B) = 50 - 44 = 6$$

(Ans)



Example 21: In a class of 50 students, 28 play Cricket and 36 play Hockey. Use Venn-diagram to find:

(i) how many play both the games?

(ii) how many play only Cricket?

Consider each of the students take part in at least one game.

Solution: (i) Let C denotes the students who play cricket and H denotes the students who play hockey.

$$\therefore n(C) = 28 \text{ and } n(H) = 36.$$

Total of these two numbers

$$= 28 + 36 = 64.$$

But total number of students in the class is 50 only; therefore some students must be playing both the games and the number of such students

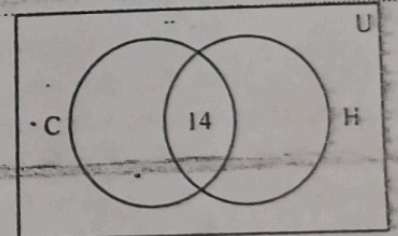
$$= 64 - 50 = 14$$

$$\text{i.e. } n(C \cap H) = 14$$

Number of students who play both the games = 14

(ii) Number of students who play only cricket

$$\begin{aligned} &= n(C) - n(C \cap H) \\ &= 28 - 14 = 14 \end{aligned}$$



Alternative Method:

Let x students play both the games.

$$\therefore \begin{aligned} &\text{Number of students who play only cricket} = 28 - x \text{ and number of students who play only} \\ &\text{hockey} = 36 - x. \end{aligned}$$

Since, the total number of students = 50

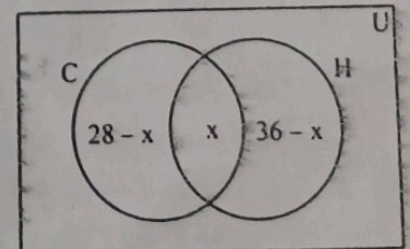
$$\therefore 28 - x + x + 36 - x = 50$$

$$\Rightarrow 64 - x = 50$$

$$\Rightarrow x = 14$$

Number of students who play both the games = 14

$$\text{Number of students who play only cricket} = 28 - 14 = 14$$



Example 22(a): A computer company has to employ 35 programmers to deal with systems programming jobs and 50 programmers to deal with application programming. Out of the programmers so employed 15 are required to deal with both type of jobs. How many programmers have to be employed?

1.18

Solution: Let A denote the set of system programmers B be the set of application programmers that have been employed. Then

$$n(A) = 35, n(B) = 50, n(A \cap B) = 15$$

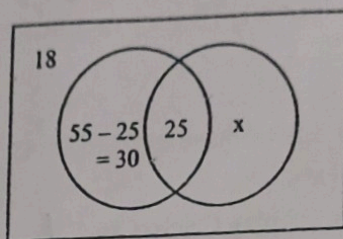
The required number of programmers to be employed is given by

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 35 + 50 - 15 \\ &= 70. \end{aligned}$$

Example 22 (b): It is found that out of 100 students, 18 can drive neither a Scooter nor a Car, while 25 can drive both these and 55 of them can drive a Scooter. How many can drive a Car.

Solution: Let A and B denote the set of students who can drive scooter and car respectively

$$n(U) - n(A \cup B) = 18 \text{ or } 100 - n(A \cup B) = 18$$



So, $n(A \cup B) = 82$. Now $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ giving
 $82 = 55 + n(B) - 25$ giving $n(B) = 52$

Example 23: Suppose that 100 of the mathematics students at a college take at least one of the languages French, German, and Russian. Also suppose:

- 60 study French
- 40 study German
- 37 study Russian
- 15 study French and German
- 20 study French and Russian
- 10 study German and Russian

Draw a Venn diagram and fill in the correct number of students in each region. Also determine the number of students who study all the three subjects.

Solution: Let F, G and R denote the sets of students studying French, German and Russian respectively. We want to find the number of students who study all three languages, and to fill in the correct number of students in each of the eight regions of the Venn-diagram shown in the figure.

We know that

$$n(F \cup G \cup R) = n(F) + n(G) + n(R) - n(F \cap G) - n(F \cap R) - n(G \cap R) + n(F \cap G \cap R)$$

Now, $n(F \cup G \cup R) = 100$, because 100 of the students study at least one of the language. Substituting the given values.

$$100 = 60 + 40 + 37 - 15 - 20 - 10 + n(F \cap G \cap R)$$

so, $n(F \cap G \cap R) = 8$, i.e. 8 students study all three languages.

We now use this result to fill in the Venn diagram. We have,

8 study all three languages

$15 - 8 = 7$ study French and German but not Russian

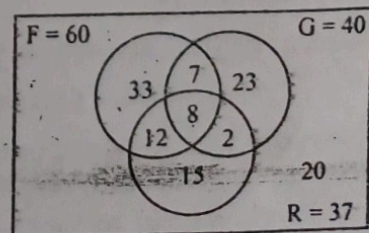
$20 - 8 = 12$ study French and Russian but not German

$10 - 8 = 2$ study German and Russian but not French

$60 - 7 - 12 - 8 = 33$ study only French

$40 - 7 - 2 - 8 = 23$ study only German

$37 - 2 - 12 - 8 = 15$ study only Russian



SET THEORY

$130 - 100 = 30$ do not study any of the languages
we see that $33 + 23 + 15 = 51$ students study exactly one of the three languages.

Example 24: Show the following sets through Venn-diagram

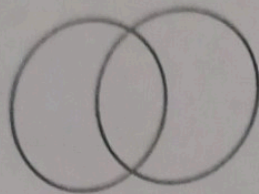
(i) $A \not\subset B$

(ii) $A \subset B$

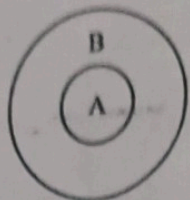
(iii) $A \not\subset B, B \subset A$

(iv) $(A \cup B) \subset (A \cup C)$ and $B \not\subset C$

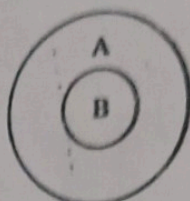
Solution:



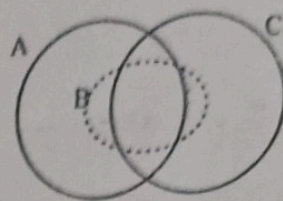
$A \not\subset B$



$A \subset B$



$A \not\subset B, B \subset A$



Example 25: In a group of 30 persons 12 are males, 9 literate and 3 male literates. Find the number of female illiterates.

Solution The Venn-diagram is as shown here. The numbers of females

$$= 30 - 12 = 18$$

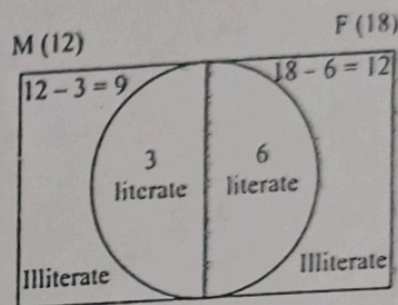
$$\text{Number of male illiterate} = 12 - 3 = 9$$

$$\text{Number of literates} = 9$$

$$\text{Number of male literate} = 3$$

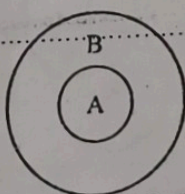
$$\text{So number of female literate} = 9 - 3 = 6$$

$$\text{Number of female illiterate} = 18 - 6 = 12$$



Example 26: Set A has 6 elements and set B has 10 elements. Find the minimum possible elements in $A \cup B$

Solution: $n(A \cup B)$ shall be minimum only if $A \subset B$ because $n(A) < n(B)$.
and then $n(A \cup B) = n(B) = 10$.



Note: If $n(B) < n(A)$, then $A \cup B$ in minimum only if $B \subset A$ and then $n(A \cup B) = n(A)$

Example 27: In an examination 70% of the students passed in Hindi, 60% in English 25% failed in both the subjects and 220 students passed in both the subjects. Find the total number of students.

Solution: Let the total number of students be x

Then, if A denotes the set of students who passed in Hindi
and B denotes the set of students who passed in English

$$n(A) = \frac{70x}{100}, n(B) = \frac{60}{100}x, n(A \cap B) = 220$$

Also $n(A \cup B) = \text{number of students who passed in Hindi or English or both Hindi and English}$
 $= \text{Total number of students who appeared at the examination} - \text{number of students who failed in Hindi and English both}$

$$= x - \frac{25}{100}x$$

By the fundamental principle of counting,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\text{or } x - \frac{25}{100}x = \frac{70x}{100} + \frac{60x}{100} - 220$$

$$\text{or } x - \left(\frac{25 + 70 + 60}{100} \right)x = -220$$

$$\text{or } x - \frac{155}{100}x = -220$$

$$\text{or } \frac{-55x}{100} = -220 \text{ giving } x = 400$$

Alternative Method

Let $x\%$ of the students pass in Hindi and English then $n(A) = 70\%$, $n(B) = 60\%$, $n(A \cap B) = x\%$

Number of students who passed in Hindi or English or both Hindi and English

$$= n(A \cup B) = 100 - 25 = 75\%$$

By fundamental counting principle,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\text{or } 75 = 70 + 60 - x$$

$$\text{or } x = 55\%$$

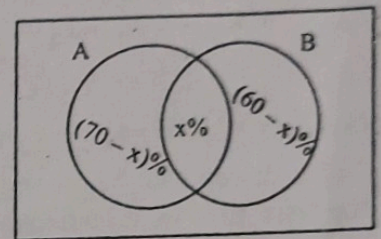
i.e., 55% students passed in both the subjects which is given as 220

Therefore 55% of total number of students = 220

$$\text{or } \frac{55}{100} \text{ of total number of students} = 220$$

$$\text{or } \text{total number of students} = \frac{220 \times 100}{55} = 400$$

Venn-diagram is as shown here



Example 28: In a survey of 100 students, it was found that 50 students liked tea, 40 liked coffee and 30 liked cold drinks. Out of these 20 liked both tea and coffee, 15 liked coffee and cold drinks and 10 liked tea and cold drinks. How many students liked all i.e. tea, coffee and cold drinks (each student liked one or more of the 3 items definitely).

Solution: Let A denote the set of students who liked tea
B denote the set of students who liked coffee
and, C denote the set of students who liked cold drinks

Then $n(A) = 50$, $n(B) = 40$, $n(C) = 30$

$n(A \cap B) = 20$, $n(B \cap C) = 15$, $n(A \cap C) = 10$

By fundamental counting principle,

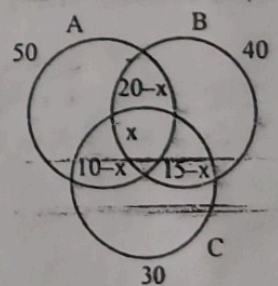
$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$\text{or } 100 = 50 + 40 + 30 - 20 - 15 - 10 + n(A \cap B \cap C)$$

$$\text{or } 100 = 120 - 45 + n(A \cap B \cap C)$$

$$\text{or } n(A \cap B \cap C) = 100 - 120 + 45 = 25$$

Venn-diagram of the problem is given here, which can also be used to solve the question.



Example 29: In a market survey opinion of the consumers regarding the following three attributes as the quality criteria to purchase tea was invited.

A - colour, B - taste, C - flavour. The following results were obtained

Number of persons interviewed = 1000

Number of persons who attribute A as criteria = 700

B	= 600	C	= 500
B and C	= 400	A and C	= 30
A and B	= 250	A, B and C	= 200

Find the number of persons who attributed (a) taste B only (b) colour A and taste B only as the criteria

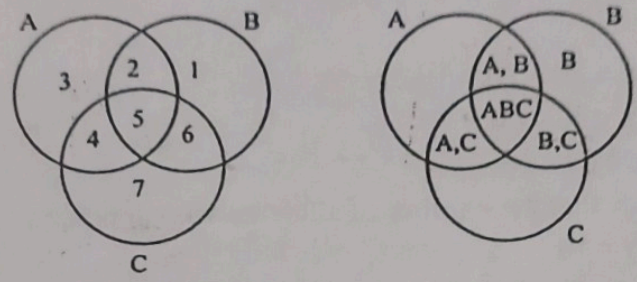
Solution: $n(B \text{ only}) = n(B) - n(A, B) - n(C, B) + n(A, B, C)$, where comma means intersection

$$= 600 - 250 - 400 + 200 = 150$$

$n(A, B \text{ only}) = n(A, B) - n(A, B, C)$

$$= 250 - 200 = 50$$

Alternative Method (Venn-diagram)



$n(B \text{ only}) = 1$

$$= (1 + 2 + 5 + 6) - (2 + 5) - (5 + 6) + 5$$
$$= n(B) - n(A, B) - n(B, C) + n(A, B, C)$$
$$= 600 - 250 - 400 + 200 = 150$$

$n(A, B) = 50$

Example 30: A statistician interviewed 1000 persons in a certain locality and found that 600 persons liked one brand of tea and 500 persons liked another brand of tea. He draws the conclusion that at least 100 persons liked both the brands of tea. Is his conclusion valid?

What would he have concluded if it is taken for granted (or is given) that each person interviewed had given his preference for either of the two or both the brands of tea?

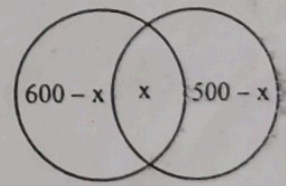
Solution: Let x be the number of persons who liked both the brands of tea. Then from venn-diagram we have "Number of preference for first brand only, for both brands and for second brand only ≤ 1000 " as some person might have not given their preferences at all

$$(600 - x) + x + (500 - x) \leq 1000$$

or $1100 - x \leq 1000$

or $x \leq -100$

or $x \geq 100$



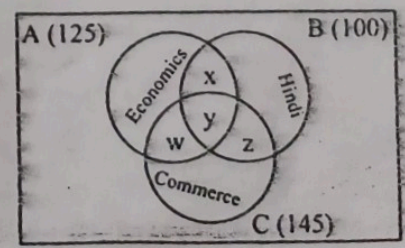
In the second case only equality sign holds
i.e. $x = 100$

Example 31: 125, 100 and 145 students appeared at post graduate examination in certain University in the subjects Economics, Hindi and Commerce respectively. 25 students appeared in Economics only, 30 in Economics and Hindi, 80 in Economics and Commerce and 60 appeared in Hindi and Commerce. Use Venn-diagram to find the number of students who appeared.

- (i) In all the three subjects
- (ii) In Hindi only
- (iii) In Commerce only

Solution: The number of students who appeared in different subjects is given as follows:

In Economics Only	= 25
In Economics & Hindi	= 30
In Economics & Commerce	= 80



In Hindi and Commerce	= 60
In Economics	= 125
In Hindi	= 100
In Commerce	= 145

Let the three circles denote the number of students who appeared in the three subjects as shown in the figure. Let the different region of the circles be denoted by w, x, y, z as shown in the figure.

$$x + y = 30 \quad \dots(1)$$

$$w + y = 80 \quad \dots(2)$$

$$y + z = 60 \quad \dots(3)$$

From the given figure, we have

$$w = 125 - 25 - (x + y) = 125 - 25 - 30 = 70$$

Hence

$$\text{from (2), } y = 80 - w = 80 - 70 = 10$$

$$\text{from (1), } x = 30 - y = 30 - 10 = 20$$

$$\text{from (3), } z = 60 - y = 60 - 10 = 50$$

From these values, we conclude that the number of students who appeared

(i) in all the three subjects = $y = 10$

(ii) in Hindi only = $100 - x - y - z = 100 - 20 - 10 - 50 = 20$

(iii) in Commerce only = $145 - w - y - z = 145 - 70 - 10 - 50 = 15$

21. SOME IMPORTANT THEOREMS ON SETS

(Use of algebraic methods as well as Venn-diagram)

Theorem 2: The operations of Union ($\cup$) and intersection ($\cap$) are commutative.

or $X \cup Y = Y \cup X$

and $X \cap Y = Y \cap X$.

Proof: To prove $X \cup Y = Y \cup X$ we have to prove that $X \cup Y \subseteq Y \cup X$

and $Y \cup X \subseteq X \cup Y$.

Let $x \in X \cup Y$, then

$$x \in X \cup Y \Rightarrow x \in X \text{ or } x \in Y$$

$$\Rightarrow x \in Y \text{ or } x \in X$$

$$\Rightarrow x \in Y \cup X.$$

So any member of $X \cup Y$ is also an element of $Y \cup X$

$$\therefore X \cup Y \subseteq Y \cup X \quad \dots (1)$$

again if $y \in Y \cup X \Rightarrow y \in Y \text{ or } y \in X$

$$\Rightarrow y \in X \text{ or } y \in Y$$

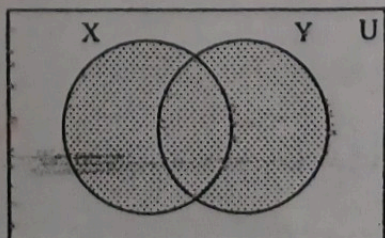
$$\Rightarrow y \in (X \cup Y)$$

$$\text{So } Y \cup X \subseteq (X \cup Y) \quad \dots(2)$$

From (1) and (2) $X \cup Y = Y \cup X$

Similarly it can also be proved that $X \cap Y = Y \cap X$.

Note: It can be proved by Venn-diagram by showing that the area representing $X \cup Y$ and $Y \cup X$ is the same as shown below



Theorem 3: The operation of Union ($\cup$) and intersection ($\cap$) are associative.

or (i) $X \cup (Y \cup Z) = (X \cup Y) \cup Z$

and (ii) $X \cap (Y \cap Z) = (X \cap Y) \cap Z$.

Proof: (i) Here also we shall prove that

$$X \cup (Y \cup Z) \subseteq (X \cup Y) \cup Z$$

and $(X \cup Y) \cup Z \subseteq X \cup (Y \cup Z)$

let $x \in X \cup (Y \cup Z) \Rightarrow x \in X \text{ or } x \in Y \cup Z$

$$\Rightarrow x \in X \text{ or } (x \in Y \text{ or } x \in Z)$$

$$\Rightarrow (x \in X \text{ or } x \in Y) \text{ or } x \in Z$$

$$\Rightarrow x \in (X \cup Y) \cup Z$$

So $X \cup (Y \cup Z) \subseteq (X \cup Y) \cup Z$... (1)

Again let $y \in (X \cup Y) \cup Z \Rightarrow y \in (X \cup Y) \text{ or } y \in Z$

$$\Rightarrow (y \in X \text{ or } y \in Y) \text{ or } y \in Z$$

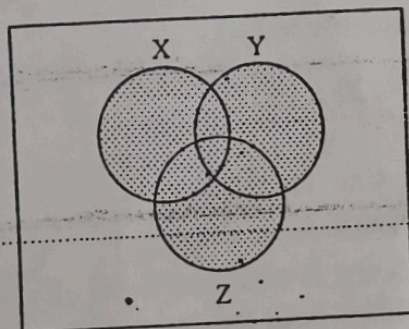
$$\Rightarrow y \in X \text{ or } (y \in Y \text{ or } y \in Z)$$

$$\Rightarrow y \in X \text{ or } y \in (Y \cup Z)$$

$$\Rightarrow y \in X \cup (Y \cup Z) \Rightarrow (X \cup Y) \cup Z \subseteq X \cup (Y \cup Z)$$

So $(X \cup Y) \cup Z \subseteq X \cup (Y \cup Z)$

Note: It can also be proved by using Venn-diagram by showing that both the sides are represented by the same area as shown in the figure



Similarly it can also be proved that $X \cap (Y \cap Z) = (X \cap Y) \cap Z$

Theorem 4: The operation of Union ($\cup$) is distributive over the operation of intersection ($\cap$)

or $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$

Proof: Let $x \in X \cup (Y \cap Z)$

$$\Rightarrow x \in X \text{ or } x \in (Y \cap Z)$$

$$\Rightarrow x \in X \text{ or } (x \in Y \text{ and } x \in Z)$$

$$\Rightarrow (x \in X \text{ or } x \in Y) \text{ and } (x \in X \text{ or } x \in Z)$$

$$\Rightarrow x \in (X \cup Y) \text{ and } x \in (X \cup Z)$$

$$\Rightarrow x \in (X \cup Y) \cap (X \cup Z)$$

So $X \cup (Y \cap Z) \subseteq (X \cup Y) \cap (X \cup Z)$... (1)

Similarly let $y \in (X \cup Y) \cap (X \cup Z)$

$$\Rightarrow (y \in X \text{ or } y \in Y) \text{ and } (y \in X \text{ or } y \in Z)$$

$$\Rightarrow y \in X \text{ or } y \in (Y \cap Z)$$

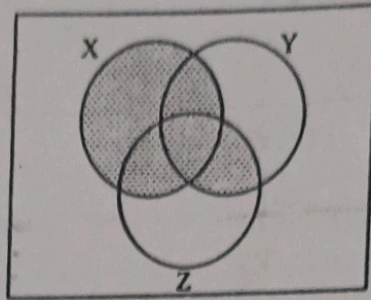
$$\Rightarrow y \in X \cup (Y \cap Z)$$

So $(X \cup Y) \cap (X \cup Z) \subseteq X \cup (Y \cap Z)$... (2)

From (1) and (2)

$$X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z) \text{ or } X \cup (Y \cap Z) \Leftrightarrow (X \cup Y) \cap (X \cup Z)$$

Note: It can also be proved by using Venn-diagram by showing that both the sides i.e. $X \cup (Y \cap Z)$ and $(X \cup Y) \cap (X \cup Z)$ are represented by the same area as shown below:



$$X \cup (Y \cap Z) \text{ or } (X \cup Y) \cap (X \cup Z)$$

Theorem 5: The operation intersection ($\cap$) is distributive over the operation Union ($\cup$)

or $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$

Proof: Let $x \in X \cap (Y \cup Z)$

$$\begin{aligned} \Rightarrow & x \in X \text{ and } (x \in Y \text{ or } x \in Z) \\ \Rightarrow & (x \in X \text{ and } x \in Y) \text{ or } (x \in X \text{ and } x \in Z) \\ \Rightarrow & x \in (X \cap Y) \text{ or } x \in (X \cap Z) \\ \Rightarrow & x \in (X \cap Y) \cup (X \cap Z) \\ \text{so} & X \cap (Y \cup Z) \subseteq (X \cap Y) \cup (X \cap Z) \end{aligned}$$

... (1)

again let $y \in (X \cap Y) \cup (X \cap Z)$

$$\begin{aligned} \Rightarrow & (y \in X \cap Y) \text{ or } (y \in X \cap Z) \\ \Rightarrow & (y \in X \text{ and } y \in Y) \text{ or } (y \in X \text{ and } y \in Z) \\ \Rightarrow & y \in X \text{ and } (y \in Y \text{ or } y \in Z) \\ \Rightarrow & y \in X \text{ and } y \in (Y \cup Z) \\ \Rightarrow & y \in X \cap (Y \cup Z) \end{aligned}$$

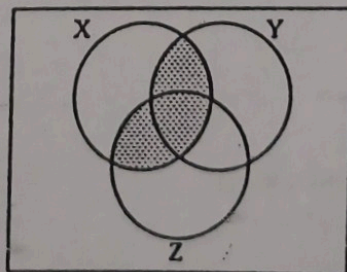
$$\text{So } (X \cap Y) \cup (X \cap Z) \subseteq X \cap (Y \cup Z)$$

... (2)

From (1) and (2)

$$X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$$

Note: It can also be proved using Venn-diagram by showing that both the sides are represented by the same area as shown below:



$$X \cap (Y \cup Z) \text{ or } (X \cap Y) \cup (X \cap Z)$$

Theorem 6: To prove that

$$(i) X \cup (Y - Z) = (X \cup Y) - (Z - X)$$

$$\text{and (ii) } X \cap (Y - Z) = (X \cap Y) - (X \cap Z)$$

$$\text{or } = (X \cap Y) - (Y \cap Z)$$

$$\text{or } = (X \cap Y) - Z$$

SET THEORY

Proof: (i)

Let $x \in X \cup (Y - Z)$

$$\begin{aligned}
 &\Rightarrow x \in X \text{ or } x \in (Y - Z) \\
 &\Rightarrow x \in X \text{ or } (x \in Y \text{ and } x \notin Z) \\
 &\Rightarrow (x \in X \text{ or } x \in Y) \text{ and } (x \in X \text{ or } x \notin Z) \\
 &\Rightarrow x \in (X \cup Y) \text{ and } x \in (X \cup Z') \\
 &\Rightarrow x \in (X \cup Y) \text{ and } x \in (X - Z) \\
 &\Rightarrow x \in (X \cup Y) \text{ and } x \notin (Z - X) \\
 &\Rightarrow x \in (X \cup Y) - (Z - X) \\
 &X \cup (Y - Z) \subseteq (X \cup Y) - (Z - X) \quad \dots(1)
 \end{aligned}$$

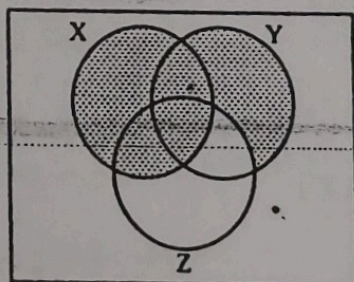
Again let $y \in (X \cup Y) - (Z - X)$

$$\begin{aligned}
 &\Rightarrow y \in (X \cup Y) \text{ and } y \notin (Z - X) \\
 &\Rightarrow (y \in X \text{ or } y \in Y) \text{ and } (y \notin Z \text{ and } y \in X) \\
 &\Rightarrow y \in X \text{ or } (y \in Y \text{ and } y \notin Z) \\
 &\Rightarrow y \in X \text{ or } y \in (Y - Z) \\
 &\Rightarrow y \in X \cup (Y - Z) \\
 &\Rightarrow (X \cup Y) - (Z - X) \subseteq X \cup (Y - Z) \quad \dots(2)
 \end{aligned}$$

From (1) and (2) we have

$$X \cup (Y - Z) = (\text{or } \Leftrightarrow) (X \cup Y) - (Z - X)$$

Note: It can also be proved, using Venn-diagram by showing that both the sides are represented by the same area as shown below.



$$X \cup (Y - Z) \text{ or } (X \cup Y) - (X \cap Z) \text{ or } (X \cup Y) - (Z - X)$$

(ii) Let $x \in X \cap (Y - Z) \Rightarrow x \in X$ and $x \in (Y - Z)$

$$\begin{aligned}
 &\Rightarrow x \in X \text{ and } (x \in Y \text{ and } x \notin Z) \\
 &\Rightarrow (x \in X \text{ and } x \in Y) \text{ and } (x \in X \text{ and } x \notin Z) \\
 &\Rightarrow x \in (X \cap Y) \text{ and } x \notin (X \cap Z) \\
 &\Rightarrow x \in (X \cap Y) - (X \cap Z)
 \end{aligned}$$

$$\text{So } X \cap (Y - Z) \subseteq (X \cap Y) - (X \cap Z) \quad \dots(1)$$

again let $y \in (X \cap Y) - (X \cap Z)$

$$\begin{aligned}
 &\Rightarrow y \in (X \cap Y) \text{ and } y \notin (X \cap Z) \\
 &\Rightarrow (y \in X \text{ and } y \in Y) \text{ and } (y \notin X \text{ or } y \notin Z) \text{ by result 11 (b) of section 20.1} \\
 &\Rightarrow (y \in X \text{ and } y \in Y) \text{ and } (y \notin Z) \\
 &\Rightarrow y \in X \text{ and } (y \in Y \text{ and } y \notin Z) \\
 &\Rightarrow y \in X \text{ and } y \in (Y - Z) \Rightarrow y \in X \cap (Y - Z)
 \end{aligned}$$

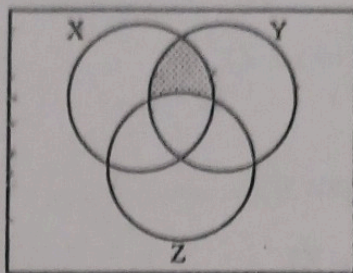
$$\text{so } (X \cap Y) - (X \cap Z) \subseteq X \cap (Y - Z) \quad \dots(3)$$

From (1) and (3)

$$X \cap (Y - Z) = (X \cap Y) - (X \cap Z)$$

From (2) we can also solve $y \in X \cap Y$ and $y \notin Z$ or $y \in (X \cap Y) - Z$

Note: It can also be proved, using Venn-diagram by showing that both the sides are represented by the same shaded area as shown below:



$$X \cap (Y - Z) \text{ or } (X \cap Y) - (Y \cap Z) \text{ or } (X \cap Y) - (X \cap Z) \text{ or } (X \cap Y) - Z$$

Theorem 7: To prove

De-Morgan's law for two sets

$$(i) (X \cup Y)' = X' \cap Y'$$

$$\text{and } (ii) (X \cap Y)' = X' \cup Y'$$

Proof: (i) Let $x \in (X \cup Y)'$

$$x \notin (X \cup Y)$$

$$x \notin X \text{ and } x \notin Y \text{ by result 11 (a).}$$

$$x \in X' \text{ and } x \in Y'$$

$$x \in X' \cap Y' \text{ for all } x \in (X \cup Y)'$$

$$(X \cup Y)' \subseteq X' \cap Y'$$

... (1)

$$\text{again let } y \in X' \cap Y'$$

$$\Rightarrow y \in X' \text{ and } y \in Y'$$

$$\Rightarrow y \notin X \text{ and } y \notin Y$$

$$\Rightarrow y \notin (X \cup Y)$$

$$\Rightarrow y \in (X \cup Y)' \text{ for all } y \in (X' \cap Y')$$

$$\text{so } X' \cap Y' \subseteq (X \cup Y)'$$

... (2)

$$\text{from (1) and (2) } (X \cup Y)' = X' \cap Y'$$

Note: It can also be proved, using Venn-diagram, by showing that both the sides are represented by the same area

$$(ii) \text{ Let } x \in (X \cap Y)' \Rightarrow x \notin X \cap Y$$

$$\Rightarrow x \notin X \text{ or } x \notin Y$$

$$\Rightarrow x \in X' \text{ or } x \in Y'$$

$$\Rightarrow x \in X' \cup Y'$$

$$\text{So } (X \cap Y)' \subseteq X' \cup Y'$$

$$\text{Again let } y \in X' \cup Y' \Rightarrow y \in X' \text{ or } y \in Y'$$

$$\Rightarrow y \notin X \text{ or } y \notin Y$$

$$\Rightarrow y \notin X \cap Y$$

$$\Rightarrow y \in (X \cap Y)'$$

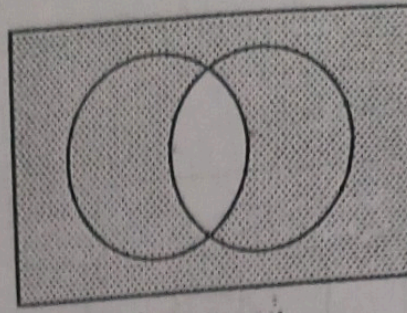
$$\text{So } X' \cup Y' \subseteq (X \cap Y)'$$

... (2)

From (1) and (2)

$$(X \cap Y)' = X' \cup Y'$$

Note: It can also be proved, using Venn-diagram, by showing that both the sides are represented by the same area as shown in the figure



$$(X \cap Y)' \text{ or } X' \cup Y'$$

Theorem 8: To prove that

$$(i) X - (Y \cap Z) = (X - Y) \cup (X - Z)$$

$$\text{and } (ii) X - (Y \cup Z) = (X - Y) \cap (X - Z)$$

[These are the alternate form of De Morgan's law]

Proof: (i) To prove both sides equal to each other we shall have to prove that set on each side is a sub set of the set on the other side

$$\text{i.e. } X - (Y \cap Z) \subseteq (X - Y) \cup (X - Z)$$

$$\text{and } (X - Y) \cup (X - Z) \subseteq X - (Y \cap Z)$$

$$\text{Let } x \in X - (Y \cap Z)$$

$$\Rightarrow x \in X \text{ and } x \notin (Y \cap Z)$$

$$\Rightarrow x \in X \text{ and } (x \notin Y \text{ or } x \notin Z)$$

$$\Rightarrow (x \in X \text{ and } x \notin Y) \text{ or } (x \in X \text{ and } x \notin Z)$$

$$\Rightarrow x \in (X - Y) \cup (X - Z)$$

$$\text{So } X - (Y \cap Z) \subseteq (X - Y) \cup (X - Z) \quad \dots (1)$$

$$\text{Again let } y \in (X - Y) \cup (X - Z)$$

$$\Rightarrow y \in (X - Y) \text{ or } y \in (X - Z)$$

$$\Rightarrow (y \in X \text{ and } y \notin Y) \text{ or } (y \in X \text{ and } y \notin Z)$$

$$\Rightarrow y \in X \text{ and } (y \notin Y \text{ or } y \notin Z)$$

$$\Rightarrow y \in X \text{ and } y \notin Y \cap Z$$

$$\Rightarrow y \in X - (Y \cap Z)$$

$$\text{So } (X - Y) \cup (X - Z) \subseteq X - (Y \cap Z) \quad \dots (2)$$

From (1) and (2),

$$X - (Y \cap Z) = (X - Y) \cup (X - Z).$$

We can write the above proof in a short form in the following alternative way:

$$\text{Let } x \in X - (Y \cap Z)$$

$$\Leftrightarrow x \in X \text{ and } x \notin Y \cap Z$$

$$\Leftrightarrow x \in X \text{ and } (x \notin Y \text{ or } x \notin Z)$$

$$\Leftrightarrow (x \in X \text{ and } x \notin Y) \text{ or } (x \in X \text{ and } x \notin Z)$$

$$\Leftrightarrow x \in (X - Y) \text{ or } x \in (X - Z)$$

$$\Leftrightarrow x \in (X - Y) \cup (X - Z)$$

$$\text{Thus, } x \in X - (Y \cap Z) \Leftrightarrow x \in (X - Y) \cup (X - Z)$$

$$\text{Therefore, } X - (Y \cap Z) = (X - Y) \cup (X - Z)$$

Note: This procedure can be followed in other questions also.

$$(ii) \text{ Let } x \in X - (Y \cup Z)$$

$$\Rightarrow x \in X \text{ and } x \notin (Y \cup Z)$$

$$\Rightarrow x \in X \text{ and } (x \notin Y \text{ and } x \notin Z)$$

$$\Rightarrow (x \in X \text{ and } x \notin Y) \text{ and } (x \in X \text{ and } x \notin Z)$$

$$\Rightarrow x \in (X - Y) \text{ and } x \in (X - Z)$$

$\Rightarrow x \in (X - Y) \cap (X - Z)$
 So $X - (Y \cup Z) \subseteq (X - Y) \cap (X - Z)$
 Again let $y \in (X - Y) \cap (X - Z)$
 $\Rightarrow y \in (X - Y)$ and $y \in (X - Z)$
 $\Rightarrow (y \in X \text{ and } y \notin Y) \text{ and } (y \in X \text{ and } y \notin Z)$
 $\Rightarrow y \in X \text{ and } (y \in Y \text{ and } y \notin Z)$
 $\Rightarrow y \in X \text{ and } y \notin (Y \cup Z)$
 $\Rightarrow y \in X - (Y \cup Z)$
 So $(X - Y) \cap (X - Z) \subseteq X - (Y \cup Z)$
 From (1) and (2) we have $X - (Y \cup Z) = (X - Y) \cap (X - Z)$

... (1)

... (2)

22. SOME MORE RESULTS ON SET THEORY

(a) To prove

If $A \subseteq B$ and $B \subseteq C$; then $A \subseteq C$

Proof: Let $x \in A \Rightarrow x \in B$ (as $A \subseteq B$)

$\Rightarrow x \in C$ (as $B \subseteq C$)

So any element of A is also an element of C . Hence $A \subseteq C$.

(b) To prove $A \subseteq A \cup B$ and $B \subseteq A \cup B$

Proof: Let $x \in A \Rightarrow x \in A \cup B$

Hence $A \subseteq A \cup B$

Similarly $B \subseteq A \cup B$

(c) To prove

(i) $A \cup A = A$

(ii) $A \cup \emptyset = A$

Proof: (i) We know $A \subseteq A \cup A$

Also $x \in A \cup A \Rightarrow x \in A$

So $A \cup A \subseteq A$

Hence $A \cup A = A$

(ii) $A \subseteq A \cup \emptyset$

Also $x \in A \cup \emptyset \Rightarrow x \in A \text{ or } x \in \emptyset$

$\Rightarrow x \in A$

So $A \cup \emptyset \subseteq A$

Hence $A \cup \emptyset = A$

(d) To prove

If $A \subseteq B$ then $A \cup B = B$

Proof: $B \subseteq A \cup B$

Also $x \in A \cup B \Rightarrow x \in A \text{ or } x \in B$

$\Rightarrow x \in B$ as $A \subseteq B$

So $A \cup B \subseteq B$

Hence $A \cup B = B$

(e) To prove

If $A \subseteq B$ then $A \cap B = A$

Proof: Let $x \in A \cap B \Rightarrow x \in A \text{ and } x \in B$

$\Rightarrow x \in A$ as $A \subseteq B$

So $A \cap B \subseteq A$

Again let $y \in A \Rightarrow y \in B$ (as $A \subseteq B$)

... (1)

SET THEORY

so $y \in A \cap B$.

i.e. $A \subseteq A \cap B$

From (1) and (2), $A \cap B = A$

(f) To prove $(A')' = A$

Proof: $x \in A \Rightarrow x \notin A' \Rightarrow x \in (A')'$

so $A \subseteq (A')'$

Again let $y \in (A')' \Rightarrow y \notin A' \Rightarrow y \in A$

So $(A')' \subseteq A$

From (1) and (2) $(A')' = A$

(g) To prove

(i) $\emptyset' = U$

(ii) $U' = \emptyset$

Proof: Let $x \in \emptyset' \Rightarrow x \notin \emptyset \Rightarrow x \in U$

So $\emptyset' \subseteq U$

Again let $y \in U \Rightarrow y \notin \emptyset \Rightarrow y \in \emptyset'$

So $U \subseteq \emptyset'$

From (1) and (2)

$\emptyset' = U$.

(ii) Let $x \in U' \Rightarrow x \notin U \Rightarrow x \in \emptyset$

So $U' \subseteq \emptyset$

Again let $y \in \emptyset \Rightarrow y \notin U \Rightarrow y \in U'$

So $\emptyset \subseteq U'$

From (1) And (2) $U' = \emptyset$

(h) If A and B are sets, then prove that $A - B$, $A \cap B$ and $B - A$ are pair wise disjoint.

Solution: First we shall prove that $A - B$ and $A \cap B$ are disjoint

i.e., $(A - B) \cap (A \cap B) = \emptyset$

We have $x \in (A - B) \cap (A \cap B) \Rightarrow x \in (A - B)$ and $x \in (A \cap B)$

$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } (x \in A \text{ and } x \in B)$.

But there is no element x which satisfies both $x \notin B$ and $x \in B$.

Therefore there is no element in the set $(A - B) \cap (A \cap B)$

i.e., $(A - B) \cap (A \cap B) = \emptyset$

Now we shall prove that $B - A$ and $A \cap B$ are disjoint

i.e., $(B - A) \cap (A \cap B) = \emptyset$.

We have $x \in (B - A) \cap (A \cap B)$

$\Rightarrow x \in (B - A)$ and $x \in (A \cap B)$

$\Rightarrow (x \in B \text{ and } x \notin A) \text{ and } (x \in A \text{ and } x \in B)$.

But there is no element x such that $x \notin A$ and $x \in A$.

Therefore there is no element in the set

$(B - A) \cap (A \cap B)$ i.e., $(B - A) \cap (A \cap B) = \emptyset$

Finally we shall prove that

$(A - B) \cap (B - A) = \emptyset$

We have $x \in (A - B) \cap (B - A)$

$\Rightarrow x \in A - B$ and $x \in B - A$

$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } (x \in B \text{ and } x \notin A)$.

But there is no element x which satisfies both $x \notin B$ and $x \in B$ or which satisfies both $x \in A$ and $x \notin A$.

Therefore $(A - B) \cap (B - A)$ has no element i.e.,

$(A - B) \cap (B - A) = \emptyset$.

(i) Prove that $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$.

Solution: We have $x \in (A - B) \cup (B - A)$

$$\begin{aligned} \Leftrightarrow & x \in (A - B) \text{ or } x \in (B - A) \\ \Leftrightarrow & (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A) \\ \Leftrightarrow & [(x \in A \text{ and } x \notin B) \text{ or } x \in B] \text{ and } [(x \in A \text{ and } x \notin B) \text{ or } x \notin A] \\ \Leftrightarrow & [x \in B \text{ or } (x \in A \text{ and } x \notin B)] \text{ and } [x \notin A \text{ or } (x \in A \text{ and } x \notin B)] \\ \Leftrightarrow & [(x \in B \text{ or } x \in A) \text{ and } (x \in B \text{ or } x \notin B)] \text{ and } [(x \notin A \text{ or } x \in A) \text{ and } (x \notin A \text{ or } x \notin B)] \\ \Leftrightarrow & x \in A \cup B \text{ and } x \notin A \cap B \Leftrightarrow x \in (A \cup B) - (A \cap B). \end{aligned}$$

Hence $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$.

(j) If $A \Delta B = (A - B) \cup (B - A)$, show that

$$A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C).$$

Solution: We have $A \cap (B \Delta C)$

$$\begin{aligned} &= A \cap [(B - C) \cup (C - B)] \\ &= [A \cap (B - C)] \cup [A \cap (C - B)] \\ &= [(A \cap B) - (A \cap C)] \cup [(A \cap C) - (A \cap B)] \\ &= (A \cap B) \Delta (A \cap C). \end{aligned}$$

[Distributive Law]

[by def. of Δ]

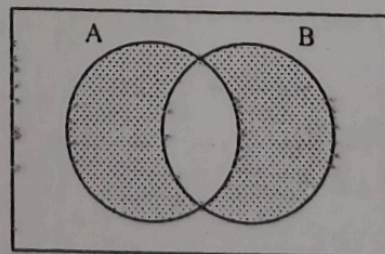
Example 32: If A and B are two sets, then prove that $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

Solution: Let $x \in (A - B) \cup (B - A)$, then

$$\begin{aligned} &x \in (A - B) \cup (B - A) \\ \Leftrightarrow &(x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A) \\ \Leftrightarrow &[(x \in A \text{ and } x \notin B) \text{ or } x \in B] \text{ and } [(x \in A \text{ and } x \notin B) \text{ or } x \notin A] \\ \Leftrightarrow &[x \in B \text{ or } (x \in A \text{ and } x \notin B)] \text{ and } [x \notin A \text{ or } (x \in A \text{ and } x \notin B)] \\ \Leftrightarrow &[(x \in B \text{ or } x \in A) \text{ and } (x \in B \text{ or } x \notin B)] \text{ and } [(x \notin A \text{ or } x \in A) \text{ and } (x \notin A \text{ or } x \notin B)] \\ &\quad \quad \quad [\text{as } x \in B \text{ or } x \notin B = U] \\ \Leftrightarrow &[x \in (A \cup B) \text{ and } x \in U] \text{ and } [x \in U \text{ and } x \notin A \cap B] \\ \Leftrightarrow &[x \in (A \cup B) \cap U] \text{ and } [x \notin (A \cap B)] \\ \Leftrightarrow &[x \in (A \cup B) \text{ and } x \notin (A \cap B)] \\ \Leftrightarrow &x \in [(A \cup B) - (A \cap B)]. \end{aligned}$$

Therefore $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$.

Note: This question can be proved easily by Venn-diagram. Each side represents the shaded area as shown below.



Example 33(a): Prove that powerset $(A \cap B) = \text{powerset}(A) \cap \text{powerset}(B)$

[UPTU., MCA., 2004-05]

Solution: Let $x \in \text{powerset}(A \cap B)$

$$\begin{aligned} \Rightarrow & x \subseteq A \cap B \\ \Rightarrow & x \subseteq A \text{ and } x \subseteq B \\ \Rightarrow & x \in \text{powerset}(A) \text{ and } x \in \text{powerset}(B) \\ \Rightarrow & x \in \text{powerset}(A) \cap \text{powerset}(B) \end{aligned}$$

...(1)

...(2)

SET THEORY

Therefore from (1) and (2)

$$\text{powerset}(A \cap B) \subseteq \overline{\text{powerset}(A)} \cap \text{powerset}(B)$$

Similarly we can prove that

$$\text{powerset}(A) \cap \text{powerset}(B) \subseteq \text{powerset}(A \cap B)$$

From (3) and (4) it is concluded that

$$\text{powerset}(A \cap B) = \text{powerset}(A) \cap \text{powerset}(B).$$

Example 33 (b): Show that for any two sets A and B.

$$A - (A \cap B) = A - B$$

Solution: Let $x \in A - (A \cap B)$

$$\Rightarrow x \in A \text{ and } (x \notin A \cap B)$$

$$\Rightarrow x \in A \text{ and } (x \notin A \text{ or } x \notin B)$$

$$\Rightarrow (x \in A \text{ and } x \notin A) \text{ or } (x \in A \text{ and } x \notin B)$$

$$\Rightarrow (x \in \phi) \text{ or } (x \in A - B) \Rightarrow x \in \phi \cup (A - B)$$

$$\Rightarrow x \in A - B$$

From (1) and (2), we have

$$A - (A \cap B) \subseteq A - B$$

Again let

$$x \in A - B$$

$$\Rightarrow x \in A \text{ and } x \notin B$$

$$\Rightarrow x \in A \text{ and } x \notin A \cap B \text{ (as } x \notin B)$$

$$\Rightarrow x \in A - (A \cap B)$$

from (4) and (5), we have

$$(A - B) \subseteq A - (A \cap B)$$

From (3) and (6), we have

$$(A - B) = A - (A \cap B)$$

[UPTU., B. Tech., 2006-07]

23. CARTESIAN PRODUCT OF SETS

Before explaining Cartesian Product of sets, we shall define ordered pair

Ordered Pair

An ordered pair, generally denoted by (x, y) , is a pair of elements x and y of two sets, which is ordered in the sense such that

$$(x, y) \neq (y, x) \text{ when } x \neq y$$

x and y are known as the first and second coordinates of the ordered pair (x, y) . The ordered pairs $(3, 5)$ and $(5, 3)$ are different because they represent different point in the coordinate plane.

Cartesian Product of two sets

The Cartesian product of two sets A and B is the set of all those ordered pairs whose first coordinate is an element of set A and the second coordinate is an element of set B . It is denoted by $A \times B$ and is read as 'A cross B' or product set of A and B .

$$\text{Thus } A \times B = \{(x, y) : x \in A \text{ and } y \in B\}$$

$$\text{Similarly } B \times A = \{(x, y) : x \in B \text{ and } y \in A\}$$

$$\text{Therefore } A \times B \neq B \times A$$

Illustration: If $A = \{2, 3, 5\}$ and $B = \{3, 4\}$, then

$$A \times B = \{(2, 3), (2, 4), (3, 3), (3, 4), (5, 3), (5, 4)\}$$

and

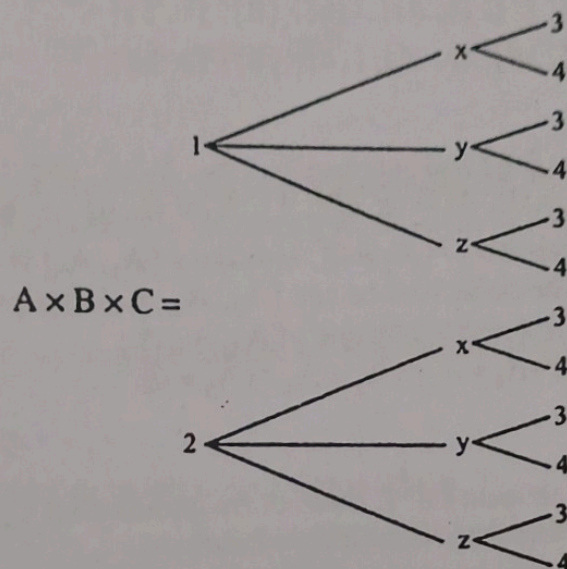
$$B \times A = \{(3, 2), (3, 3), (3, 5), (4, 2), (4, 3), (4, 5)\}$$

In a similar way, the Cartesian product for n sets $A_1, A_2, \dots, A_n$ is defined as

$$A_1 \times A_2 \times \dots \times A_n = \{(x_1, x_2, \dots, x_n) : x_1 \in A_1, x_2 \in A_2, \dots, x_n \in A_n\}$$

Example 34: If $A = \{1, 2\}$, $B = \{x, y, z\}$, $C = \{3, 4\}$, find $A \times B \times C$.

Solution:



$$= \{(1, x, 3), (1, x, 4), (1, y, 3), (1, y, 4), (1, z, 3), (1, z, 4), \\ (2, x, 3), (2, x, 4), (2, y, 3), (2, y, 4), (2, z, 3), (2, z, 4)\}$$

Note 1: Let $A = \{\alpha, \beta\}$, $B = \{1, 2, 3\}$, then

$$(A \times B) = \{(\alpha, 1), (\alpha, 2), (\alpha, 3), (\beta, 1), (\beta, 2), (\beta, 3), \}$$

and

$$(B \times A) = \{(1, \alpha), (1, \beta), (2, \alpha), (2, \beta), (3, \alpha), (3, \beta)\}$$

Hence

$$(A \times B) \cap (B \times A) = \phi$$

Note 2: Similarly if $A = \phi$, $B = \{1, 2, 3\}$, we can show that

$$A \times B = \phi \text{ and } B \times A = \phi$$

Theorem 9: If A, B, C be three non empty sets, then prove that:

$$(i) A \times (B \cup C) = (A \times B) \cup (A \times C) \quad (ii) A \times (B \cap C) = (A \times B) \cap (A \times C)$$

Solution: (i) $A \times (B \cup C) = \{(x, y) : x \in A \text{ and } y \in (B \cup C)\}$ (by def. of Cartesian product of sets)

$$= \{(x, y) : x \in A \text{ and } (y \in B \text{ or } y \in C)\}$$

$$= \{(x, y) : (x \in A \text{ and } y \in B) \text{ or } (x \in A \text{ and } y \in C)\} \quad (\text{distribution of 'and' over 'or'})$$

$$= \{(x, y) : (x, y) \in A \times B \text{ or } (x, y) \in A \times C\}$$

$$= (A \times B) \cup (A \times C)$$

(ii) We can prove it as in (i) or follow the alternative method given below which can also be applied for (i)

Solution: Let $(x, y) \in A \times (B \cap C)$

Then $(x, y) \in A \times (B \cap C)$

$$\Leftrightarrow x \in A \text{ and } y \in B \cap C$$

$$\begin{aligned} \Leftrightarrow & x \in A \text{ and } (y \in B \text{ and } y \in C) \\ \Leftrightarrow & (x \in A \text{ and } y \in B) \text{ and } (x \in A \text{ and } y \in C) \\ \Leftrightarrow & (x, y) \in (A \times B) \text{ and } (x, y) \in (A \times C) \\ \Leftrightarrow & (x, y) \in (A \times B) \cap (A \times C). \end{aligned}$$

Hence, $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

Theorem 10: Prove that

$$A \times (B - C) = (A \times B) - (A \times C)$$

$$\begin{aligned} \text{Solution: } A \times (B - C) &= \{(x, y) : x \in A \text{ and } y \in (B - C)\} \\ &= \{(x, y) : x \in A \text{ and } (y \in B \text{ and } y \notin C)\} \\ &= \{(x, y) : (x \in A \text{ and } y \in B) \text{ and } (x \in A \text{ and } y \notin C)\} \\ &= \{(x, y) : (x, y) \in (A \times B) \text{ and } (x, y) \notin (A \times C)\} \\ &= (A \times B) - (A \times C). \end{aligned}$$

Theorem 11: For any sets A, B, C, D more that

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$$

Proof: Let $(a, b) \in (A \times B) \cap (C \times D)$, then

$$\begin{aligned} \Rightarrow & (a, b) \in (A \times B) \text{ and } (a, b) \in (C \times D) \\ \Rightarrow & (a \in A \text{ and } b \in B) \text{ and } (a \in C \text{ and } b \in D). \\ \Rightarrow & (a \in A \text{ and } a \in C) \text{ and } (b \in B \text{ and } b \in D) \\ \Rightarrow & a \in (A \cap C) \text{ and } b \in (B \cap D) \\ \Rightarrow & (a, b) \in (A \cap C) \times (B \cap D) \end{aligned}$$

Since it is true for every ordered pair (a, b)

$$(A \times B) \cap (C \times D) \subseteq (A \cap C) \times (B \cap D)$$

Again let $(a, b) \in (A \cap C) \times (B \cap D)$

$$\begin{aligned} \Rightarrow & a \in (A \cap C) \text{ and } b \in (B \cap D) \\ \Rightarrow & (a \in A \text{ and } a \in C) \text{ and } (b \in B \text{ and } b \in D) \\ \Rightarrow & (a \in A \text{ and } b \in B) \text{ and } (a \in C \text{ and } b \in D) \\ \Rightarrow & (a, b) \in (A \times B) \text{ and } (a, b) \in (C \times D) \\ \Rightarrow & (a, b) \in (A \times B) \cap (C \times D) \end{aligned}$$

Since it is true for every ordered pair (a, b)

$$(A \cap C) \times (B \cap D) \subseteq (A \times B) \cap (C \times D)$$

From (1) and (2), we have

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$$

Note: It is evident from this theorem that

$$(A \times B) \cap (B \times A) = (A \cap B) \times (B \cap A)$$

Example 35: For any two finite, non-empty sets A and B.

$$|A \times B| = |A| |B|$$

i.e., cardinality of $A \times B$ = (Cardinality of A) $\times$ (Cardinality of B)

Proof: Let $|A|$ or $n(A) = n_1$

$$|B| \text{ or } n(B) = n_2$$

Elements of the set $A \times B$ are (x, y) where $x \in A$ and $y \in B$

We can choose elements $x \in A$ in n_1 ways for the first position in ordered pair (x, y) and can choose $y \in B$ in n_2 ways for second position in the ordered pair (x, y) .

Therefore by multiplication principle we can choose both x and y or the ordered pair (x, y) in $n_1 \times n_2$ ways.

Thus $|A \times B| = n_1 n_2 = |A| |B|$

Example 36: Prove that If A and B are two non-empty sets having n elements in common, then $A \times B$ and $B \times A$ have n^2 elements in common.

Proof: We know that

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$$

$$\Rightarrow (A \times B) \cap (B \times A) = (A \cap B) \times (B \cap A)$$

$$\Rightarrow (A \times B) \cap (B \times A) = (A \cap B) \times (A \cap B)$$

Since $(A \cap B)$ has n common elements therefore $(A \cap B) \times (A \cap B)$ has n^2 elements in common.

$$\Rightarrow (A \times B) \cap (B \times A) = (A \cap B) \times (B \cap A) \text{ has } n^2 \text{ elements in common.}$$

Some Important Results

- (i) If A and B are any two non-empty sets, then $A \times B = B \times A \Leftrightarrow A = B$.
- (ii) If $A \subseteq B$, then $A \times A \subseteq (A \times B) \cap (B \times A)$
- (iii) If $A \subseteq B$, then $A \times C \subseteq B \times C$ for any set C .
- (iv) If $A \subseteq B$ and $C \subseteq D$, then $A \times C \subseteq B \times D$
- (v) $A \times B = A \times C \Rightarrow B = C$.

24. PARTITIONS OR QUOTIENT SETS

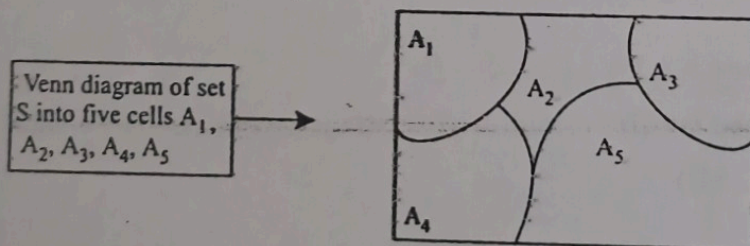
Let S be a non empty set, A partition of S is a subdivision of S into non overlapping, non empty subsets.

Or a partition of S is a collection $\{A_i\}$ of non empty subsets of S such that

(a) Each a in S belongs to one of the A_i .

(b) The sets of $\{A_i\}$ are mutually disjoint i.e. if

$$A_i \neq A_j \text{ then } A_i \cap A_j = \emptyset$$



Subsets in a partition are called Cells or blocks

Example 37: Consider following collection of subsets of

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

- (i) $\{\{1, 3, 5\}, \{2, 6\}, \{4, 8, 9\}\}$
- (ii) $\{\{1, 3, 5\}, \{2, 4, 6, 8\}, \{5, 7, 9\}\}$
- (iii) $\{\{1, 3, 5\}, \{2, 4, 6, 7\}, \{8, 9\}\}$
- (iv) $\{\{1, 3, 6\}, \{2, 8\}, \{5, 7, 9\}\}$
- (v) $\{\{1, 5, 7\}, \{2, 4, 8, 9\}, \{3, 5, 6\}\}$
- (vi) $\{\{2, 4, 5, 8\}, \{1, 9\}, \{3, 6, 7\}\}$
- (vii) $\{\{1, 2, 7\}, \{3, 5\}, \{4, 6, 8, 9\}, \{3, 5\}\}$

- (i) is not a partition of S since $7 \in S$ but does not belong to any of the subsets.
- (ii) is not a partition of S since $\{1, 3, 5\}$ and $\{5, 7, 9\}$ are not disjoint.
- (iv) is not a partition of S since $4 \in S$ but does not belong to any subset.

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(v) is not a partition of S since $5 \in S$ belongs to two distinct cells subsets.

(iii) & (vi) & (vii) each is a partition

Example 38: Find all the partition of $X = \{a, b, c, d\}$

Solution: Each partition of X contains either 1, 2, 3 or 4 distinct sets as shown below

- (1) $\{\{a, b, c, d\}\}$
- (2) (i) $\{\{a\}, \{b, c, d\}\}, \{\{b\}, \{a, c, d\}\}, \{\{c\}, \{a, b, d\}\}$
(ii) $\{\{d\}, \{a, b, c\}\}, \{\{a, b\}, \{c, d\}\}, \{\{b, c\}, \{a, d\}\}$
(iii) $\{\{a, c\}, \{b, d\}\}$
- (3) $\{\{a\}, \{b\}, \{c, d\}\}, \{\{a\}, \{c\}, \{b, d\}\}, \{\{a\}, \{d\}, \{b, c\}\}$
 $\{\{b\}, \{c\}, \{a, d\}\}, \{\{b\}, \{d\}, \{a, c\}\}, \{\{c\}, \{d\}, \{a, b\}\}$
- (4) $\{\{a\}, \{b\}, \{c\}, \{d\}\}$

Example 39: Let $A = \{a, b, c, d, e, f, g\}$ and suppose A_1, A_2, A_3, A_4 and A_5 are the subsets of A such that $A_1 = \{a, c, e, f, g\}, A_2 = \{a, b, c, d\}, A_3 = \{a, c, e, g\}, A_4 = \{b, d\}, A_5 = \{f\}$.

Here $\{A_1, A_2\}$ is not a partition because $A_1 \cap A_2 \neq \emptyset$. Similarly $\{A_1, A_5\}$ is also not a partition because $A_1 \cap A_5 \neq \emptyset$ ($f \in A_1, f \in A_5$). The collection or set $\{A_3, A_4, A_5\}$ is a partition because it satisfies condition (a) i.e. each element of A belongs to any one of A_3, A_4, A_5 .

It also satisfies condition (b) i.e. $A_3 \cap A_4 = A_3 \cap A_5 = A_4 \cap A_5 = \emptyset$

25. INDEX SET

If T is a non empty set and to each element $t \in T$, there corresponds a set C_t , then we say that each $t \in T$ indexes a C_t and the set T used in this sense is called an Index set.

The collection C of all these sets is called the indexed family of set. Symbolically $C = \{C_t : t \in T\}$

Where C_t is a set corresponding to an element $t \in T$ (for all t)

Example 40: Let the indexed set $T = \{a, b, c, d\}$ and corresponding to each element a, b, c, and d we have the sets

$$C_a = \{2, 3\},$$

$$C_b = \{2, 3, 4, 5\},$$

$$C_c = \{4, 5, 6\},$$

$$C_d = \{2, 4, 6, 8\}$$

then the indexed family of set $A = \{C_a, C_b, C_c, C_d\}$

26. ARBITRARY UNION OF SETS

If T is an index set and C_t denotes a set corresponding to each value of $t \in T$ then Union of all these sets C_t given by $\cup C_t$ is set of elements that

are contained in C_t for some $t \in T$ and is called the arbitrary union of sets. Symbolically

$$\cup_{t \in T} C_t = \{x : x \in C_t \text{ for some } t \in T\}$$

In the previous example,

$$\cup_{t \in T} C_t = C_a \cup C_b \cup C_c \cup C_d$$

$$t \in T = \{2, 3, 4, 5, 6, 8\}$$

27. FUNDAMENTAL PRODUCT OF SETS

Let there be n distinct sets $A_1, A_2, \dots, A_n$.

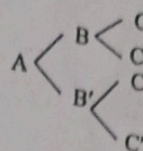
Fundamental product of these sets is

$$A_1^* \cap A_2^* \cap A_3^* \dots \cap A_n^*$$

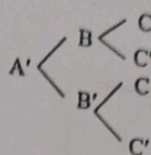
A_i^* is either A_i or complement of A_i

For 3 sets A, B and C we have $2^3 = 8$ fundamental products

$$\left. \begin{aligned} P_1 &= A \cap B \cap C \\ P_2 &= A \cap B \cap C' \\ P_3 &= A \cap B' \cap C \\ P_4 &= A \cap B' \cap C' \end{aligned} \right\}$$



$$\left. \begin{aligned} P_5 &= A' \cap B \cap C \\ P_6 &= A' \cap B \cap C' \\ P_7 &= A' \cap B' \cap C \\ P_8 &= A' \cap B' \cap C' \end{aligned} \right\}$$



28. PEANO AXIOMS

These axioms deal with generation of the set of Natural Numbers $N = \{0, 1, 2, 3, \dots\}$ (including zero) starting with a null set ϕ and the notion of successor set. A successor set of a set A denoted by A^+ (or P) is defined as follows:

$$A^+ = A \cup \{A\}$$

If ϕ is the empty set, the its successive successor sets are as given below (on the assumption that $\phi = 0$)

$$\text{Empty set} = \phi = 0$$

$$\text{Successor set of } \phi = \phi^+ = 0^+ = \phi \cup \{\phi\} = \{\phi\} = 1$$

$$\text{Successor set of } \phi^+ = \phi^+ \cup \{\phi^+\} = \{\phi, \{\phi\}\} = 2$$

$$\text{Successor set of } (\phi^+)^+ = \phi^+ \cup \{\phi^+\} \cup \{\phi^+ \cup \{\phi^+\}\} = \{\phi, \{\phi\}, \{\phi, \{\phi\}\}\} = 3$$

In this way we get the set $\{0, 1, 2, 3, \dots\}$ where each element is a successor set of the previous element except for the starting element 0 which is assumed to be present and not generated.

EXERCISES

- If $A = \{2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 7\}$ and $C = \{3, 5, 6\}$ then find:
(a) $A \cup B$, (b) $A \cup C$, (c) $B \cup C$, (d) $C \cup C$, (e) $(A \cup B) \cup C$, (f) $A \cup (B \cup C)$
[Hint: To find $(A \cup B) \cup C$, first find $(A \cup B)$ which is $\{2, 3, 4, 5, 6, 7\}$ and then $(A \cup B) \cup C = \{2, 3, 4, 5, 6, 7\} \cup \{3, 5, 6\}$]
- If $A = \{\text{Mohan, Ram, Krishan}\}$, $B = \{\text{Suresh, Mohan, Harish}\}$ and $C = \{\text{Surendra, Mohan}\}$ then find:
(a) $A \cup B$ (b) $B \cup C$ (c) $A \cup C$
- If $X = \{2, 3, 4, 5\}$, $Y = \{2, 4, 5, 8\}$, $Z = \{3, 5, 7, 8, 9\}$ then find
(a) $X \cap Y$, (b) $Y \cap Z$, (c) $(Y \cap Y) \cap Z$, (d) $X \cap (Y \cap Z)$.
[Hint: To find $(X \cap Y) \cap Z$ first evaluate $(X \cap Y)$ and then find the intersection of the set so obtained and the set Z .]
- If $X = \{a, b, c, d\}$, $Y = \{b, d, f, h\}$, $Z = \{c, d, e, f\}$ then find the following sets.
(a) $X - Y$ (b) $(Z - X)$, (c) $(Y - Z)$ (d) $Z - Z$ (e) $(Z - Y)$.
- If X and Y are two sets prove that
(a) $X - Y \subset X \cup Y$, (b) $X' - Y' = Y - X$ (c) $(X \cap Y)' = X' \cup Y'$ (d) $(X - Y) \subseteq X$.
[Hint: Let $x \in (X - Y)$ i.e. $x \in X$, $x \notin Y$;
 $\therefore x \in (X - Y)$ implies $x \in X$ etc.]
- (a) Define complement set, proper subset and intersection sets. Given examples in each case.
(b) If $B \subset A$, prove that $B \cup D \subset A \cup D$.

[Hint: Let $x \in B$, As $B \subset A$, $x \in A$ If $y \in B \cup D$, then $y \in B$ or $y \in D$
 $\therefore y \in A$ or $y \in D \therefore y \in A \cup D$]

7. Prove that $A - B$ is subset of $A \cup B$. If in particular $D = \{2, 4, 6, 8\}$ and $C = \{3, 4, 5, 6\}$ find $D - C$
8. If $A = \{a, b, c, d\}$, $B = \{f, b, d, g\}$ determine $A \cup B$, $A \cap B$, $A - B$, $B - A$, $A \cup A$
9. Define a set and a subset. If $A = \{1, 3, 2\}$ and $B = \{2, 4, 6\}$ calculate the set $A \cup B$, $A \cap B$, $A \sim B$.
10. If $A = \{1, 3, 5, 7\}$, $B = \{2, 3, 5\}$, $C = \{5, 6, 8, 9\}$
 $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ find
 (i) $(A \cup B \cup C)$ (ii) $(A \cap B \cap C)$ (iii) $(A \cup B) \cap C$ (iv) $(A \cup B) - C$ (v) $(B - C)$
11. If $A = \{a, b, c, d\}$, $B = \{b, d, e, f\}$, $C = \{b, c, g\}$, prove that
 (i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 (ii) $A - (B \cup C) = (A - B) \cap (A - C)$
12. If $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4, 6, 8\}$, $C = \{3, 5, 7\}$ find
 $A \cup (B \cap C)$, $(A \cup B) \cap C$, $A - (B - C)$, $A \cap (B \cap C)$, $(A \cap B) \cup C$
13. If $A = \{a, b, c\}$, $B = \{b, c, d, e\}$, $C = \{c, e, b\}$ find
 (i) $A \cup (B \cap C)$ (ii) $A - (B - C)$ (iii) $(A \cup B) - C$ (iv) $A \cap (B \cap C)$
14. In a group of 70 people, 37 like coffee, 52 like tea and each person likes at least one of the two drinks. How many people like both coffee and tea?
 [Hint: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$]
15. In a market survey 100 consumers were interviewed for their preference in respect of two products A and B. It was found that 64 liked product A while 67 liked product B. Assuming that every consumer gave his performance, state how many liked both A and B.
16. If $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$
 $A = \{1, 2, 5, 6\}$, $B = \{2, 3, 6, 7, 8\}$ prove that
 (i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$
17. Draw a Venn diagram to show the sets $A = \{1, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$. Shade the region representing
 (i) $A - (A \cap B)$ (ii) $A' \cap B$
18. Draw a Venn diagram to represent the relationship between the following sets.
 $U = \{\text{Quadrilaterals}\}$, $A = \{\text{Parallelograms}\}$, $B = \{\text{Rectangles}\}$, $C = \{\text{Rhombus}\}$.
 State that region in the diagram which represents the set of squares.
19. State in set language, what these shaded regions represent in the given Venn-diagrams.

